

#1 · Clinical Chemistry

What is the standard CLSI order of draw for venipuncture tubes, from first to last?

Answer

Blood culture bottles, light blue (citrate), red or gold (clot activator), green (heparin), lavender (EDTA), then gray (fluoride and oxalate). This order prevents additive carryover between tubes.

#2 · Clinical Chemistry

Which chemistry results are falsely changed when EDTA carries over into a serum tube?

Answer

Potassium rises falsely because EDTA is a potassium salt. Calcium, magnesium and alkaline phosphatase fall because EDTA binds divalent cations that the assays or enzymes need.

#3 · Clinical Chemistry

Which two unique identifiers are acceptable when identifying a patient before collection?

Answer

Full name plus date of birth or a health card number. Room and bed numbers are never used, because patients move, and the patient states their own details.

#4 · Clinical Chemistry

Why must a light blue top citrate tube be filled to the marked line?

Answer

The ratio of blood to sodium citrate must be 9 to 1. An underfilled tube has excess anticoagulant, which falsely prolongs clotting times.

#5 · Clinical Chemistry

How long should a tourniquet stay on during venipuncture, and what happens if it stays longer?

Answer

Less than one minute. Longer application causes hemoconcentration, raising protein-bound analytes such as total protein, calcium and cholesterol, and it can raise potassium.

#6 · Clinical Chemistry

What additive does a gray top tube contain and why is it used for glucose?

Answer

Sodium fluoride and potassium oxalate. Fluoride blocks glycolysis so red cells stop consuming glucose, and oxalate binds calcium to prevent clotting.

#7 · Clinical Chemistry

Where is a heel puncture performed on an infant younger than one year?

Answer

On the lateral or medial plantar surface of the heel. The central heel is avoided because bone lies close to the skin and can be damaged.

#8 · Clinical Chemistry

What is the effect of hemolysis on potassium, and which other analytes rise?

Answer

Potassium is falsely raised because red cells hold high potassium. Magnesium, phosphate, lactate dehydrogenase, aspartate aminotransferase and iron also rise.

#9 · Clinical Chemistry

What is pseudohyponatremia and which sample condition causes it?

Answer

A falsely low sodium result seen in lipemic samples measured by indirect ion-selective electrode. High lipid takes up plasma volume that the method does not count.

#10 · Clinical Chemistry

Within what time should serum or plasma be separated from cells if no gel barrier is used?

Answer

Within about two hours. Cells left in contact continue to consume glucose and leak potassium, phosphate and lactate dehydrogenase into the serum.

#11 · Clinical Chemistry

Why is whole blood for potassium never refrigerated before centrifugation?

Answer

Cold stops the sodium and potassium pump in red cells, so potassium leaks into the serum and gives a falsely high result.

#12 · Clinical Chemistry

How should a specimen for ammonia be handled?

Answer

Collect in EDTA or heparin, place it on ice and deliver it to the laboratory within about 15 minutes, since cells release ammonia and raise the value.

#13 · Clinical Chemistry

What is a trough level and when is the sample drawn?

Answer

A trough is the lowest drug concentration, drawn just before the next dose. Drawing it after a dose gives a falsely high value and may prompt an unsafe dose cut.

#14 · Clinical Chemistry

What pattern suggests contamination by a dextrose intravenous infusion in a chemistry sample?

Answer

A very high glucose with low values for other analytes because of dilution. The sample was probably drawn above the infusion and should be recollected.

#15 · Clinical Chemistry

What should an MLT check before calling a critical potassium result?

Answer

The hemolysis index, tube type, delta check and clinical context. If the sample is hemolyzed, the provider is told and an urgent recollection is requested.

#16 · Clinical Chemistry

State the Beer-Lambert law and what each term represents.

Answer

$A = abc$: absorbance equals absorptivity times path length times concentration. Within the linear range, absorbance rises in a straight line with concentration.

#17 · Clinical Chemistry

What should a technologist do with a result above the method's linearity limit?

Answer

Dilute the sample with saline, re-run it, and multiply the result by the dilution factor. A value above the measuring range is never reported without dilution.

#18 · Clinical Chemistry

A 5.0 mmol/L standard reads absorbance 0.250 and a patient reads 0.400. What is the patient concentration?

Answer

8.0 mmol/L, calculated as 0.400 divided by 0.250, multiplied by 5.0. Concentration is proportional to absorbance in the linear range.

#19 · Clinical Chemistry

Why can indirect ion-selective electrode methods give pseudohyponatremia?

Answer

The sample is diluted first, and high lipid or protein takes up plasma volume that the calculation counts as water, giving a falsely low sodium. Direct electrodes avoid this.

#20 · Clinical Chemistry

Which immunoassay format suits small molecules such as cortisol, and how does its signal behave?

Answer

The competitive format. Sample analyte competes with labelled analyte for limited antibody sites, so the signal falls as the analyte concentration rises.

#21 · Clinical Chemistry

What is the hook effect and how is it corrected?

Answer

In a sandwich immunoassay, extremely high analyte saturates both antibodies separately, so the sandwich fails and the result is falsely low. Dilute the sample and repeat.

#22 · Clinical Chemistry

Why does chemiluminescence give high sensitivity in immunoassays?

Answer

The label, such as acridinium ester, makes light from a chemical reaction without an excitation source, so background signal is very low and small amounts can be detected.

#23 · Clinical Chemistry

What does a sharp dense band in the gamma region of serum protein electrophoresis suggest?

Answer

A monoclonal protein, such as in myeloma. The next step is immunofixation to identify the heavy and light chain type.

#24 · Clinical Chemistry

How does LC-MS/MS identify a molecule?

Answer

It separates compounds by liquid chromatography, ionizes them, and identifies them by mass-to-charge ratio and fragments. It is more specific than immunoassay for steroids and drugs.

#25 · Clinical Chemistry

What is random access on a chemistry autoanalyzer?

Answer

Any test can be run on any sample in any order, because each test uses its own reaction vessel. Stat samples can be placed ahead of routine work.

#26 · Clinical Chemistry

What are two disadvantages of point-of-care testing compared with central laboratory testing?

Answer

Lower precision and more chance of operator error. Device-specific interferences also occur, for example hematocrit effects on glucose meters.

#27 · Clinical Chemistry

When must an analyzer be recalibrated?

Answer

With a new reagent lot, after major maintenance, when control shifts cannot be otherwise corrected, or when calibration expires. Controls must pass before patient results are reported.

#28 · Clinical Chemistry

Which Westgard rules point to systematic error?

Answer

The 2-2s, 4-1s and 10x rules. The 1-3s and R-4s rules usually point to random error, and the 1-2s rule is only a warning.

#29 · Clinical Chemistry

What is the coefficient of variation and how is it calculated?

Answer

CV is the standard deviation divided by the mean, multiplied by 100. It expresses precision as a percentage so that methods at different concentrations can be compared.

#30 · Clinical Chemistry

What does a 95% reference interval mean for healthy people?

Answer

About 95% of healthy people fall inside it and about 5% fall outside. A result outside the interval is therefore not automatically disease.

#31 · Clinical Chemistry

What is the diagnostic fasting plasma glucose value for diabetes in Canada?

Answer

A fasting plasma glucose of 7.0 mmol/L or higher, with fasting defined as no caloric intake for at least 8 hours.

#32 · Clinical Chemistry

Why does unseparated whole blood give a falsely low glucose result?

Answer

Red and white cells keep using glucose by glycolysis in the tube, lowering it roughly 5 to 7 percent per hour, faster if the white count is very high.

#33 · Clinical Chemistry

How does sodium fluoride preserve glucose in a collection tube?

Answer

It inhibits the enzyme enolase in the glycolytic pathway, though it acts slowly, so prompt separation is still advised. It is combined with potassium oxalate as the anticoagulant.

#34 · Clinical Chemistry

What period of average glucose does HbA1c reflect, and why?

Answer

About two to three months, because red cells live roughly 120 days and glucose attaches permanently to hemoglobin during that time.

#35 · Clinical Chemistry

Which conditions make HbA1c unreliable as a measure of glycemic control?

Answer

Anything that changes red cell lifespan, such as hemolytic anemia, blood loss or transfusion, and some hemoglobin variants like S, C, E or D that interfere with certain methods.

#36 · Clinical Chemistry

Which ketone does the nitroprusside urine test fail to detect, and why does that matter in DKA?

Answer

It misses beta-hydroxybutyrate, the main ketone in diabetic ketoacidosis, so blood beta-hydroxybutyrate measurement is more reliable for monitoring.

#37 · Clinical Chemistry

Write the Friedewald equation for LDL cholesterol in SI units.

Answer

LDL-C equals total cholesterol minus HDL-C minus triglycerides divided by 2.2, with all values in mmol/L.

#38 · Clinical Chemistry

When is the Friedewald calculation not valid?

Answer

When triglycerides exceed 4.5 mmol/L, or in dysbetalipoproteinemia. A direct LDL-C measurement should then be performed or the calculation not reported.

#39 · Clinical Chemistry

Why is serum rather than plasma used for serum protein electrophoresis?

Answer

Plasma contains fibrinogen, which creates an extra band between the beta and gamma regions that can be mistaken for a monoclonal protein.

#40 · Clinical Chemistry

What does a narrow, dense spike in the gamma region on SPE suggest, and what confirms it?

Answer

A monoclonal protein, as in multiple myeloma or related disorders. Immunofixation confirms it and identifies the heavy and light chain type.

#41 · Clinical Chemistry

Why is creatinine preferred over urea as a marker of kidney function?

Answer

Creatinine production is steady in proportion to muscle mass, while urea varies with protein intake, hydration, bleeding and liver function.

#42 · Clinical Chemistry

What inputs does the CKD-EPI eGFR equation use, and when is it not valid?

Answer

Serum creatinine, age and sex. It is not valid when creatinine is not in a steady state, such as in acute kidney injury.

#43 · Clinical Chemistry

Why does conjugated bilirubin appear in urine but unconjugated bilirubin does not?

Answer

Conjugated bilirubin is water soluble and passes the kidney, while unconjugated bilirubin is bound to albumin and is not filtered.

#44 · Clinical Chemistry

Which liver enzyme is more specific for hepatocyte injury, ALT or AST, and why?

Answer

ALT is more specific because it is found mainly in liver, while AST is also found in heart, skeletal muscle and red cells.

#45 · Clinical Chemistry

How does a raised GGT help interpret a raised ALP?

Answer

A raised GGT alongside raised ALP points to a liver or bile duct source, while a normal GGT suggests bone or another source.

#46 · Clinical Chemistry

Name four pre-analytical causes of falsely high potassium.

Answer

Hemolysis, delayed separation of the sample, fist clenching or prolonged tourniquet use, and refrigeration of whole blood before centrifugation. EDTA contamination also adds potassium.

#47 · Clinical Chemistry

Which pattern suggests EDTA contamination of a serum sample?

Answer

A sudden high potassium together with low calcium, and often low magnesium and alkaline phosphatase, since EDTA adds potassium and binds divalent cations.

#48 · Clinical Chemistry

Why can indirect ISE give a falsely low sodium in a very lipemic sample?

Answer

Lipid or protein reduces the water fraction of plasma, so diluted-sample measurement underestimates sodium. Direct ISE avoids this error, called pseudohyponatremia.

#49 · Clinical Chemistry

Write the albumin correction formula for total calcium in mmol/L.

Answer

Corrected calcium equals measured total calcium plus 0.02 times (40 minus albumin in g/L). It is only an estimate, and ionized calcium is better in critical illness.

#50 · Clinical Chemistry

Why must ionized calcium samples be collected anaerobically?

Answer

Loss of carbon dioxide raises pH, which increases calcium binding to protein and lowers the ionized calcium result.

#51 · Clinical Chemistry

How is the anion gap calculated, and what is an example of a high result?

Answer

Sodium minus the sum of chloride and bicarbonate. For 140, 100 and 12 mmol/L the gap is 28, which is high and suggests an added acid.

#52 · Clinical Chemistry

Name causes of a high anion gap metabolic acidosis.

Answer

Lactic acidosis, diabetic ketoacidosis, uremia, and poisoning with methanol, ethylene glycol or salicylate.

#53 · Clinical Chemistry

How is the osmolal gap calculated and what does a gap above 10 suggest?

Answer

Measured osmolality minus calculated osmolality, where calculated is 2 times sodium plus glucose plus urea. A gap above about 10 suggests toxic alcohols or other unmeasured solutes.

#54 · Clinical Chemistry

What are the expected findings in respiratory acidosis?

Answer

Low pH with high pCO₂ from hypoventilation, and the kidneys compensate over time by raising bicarbonate.

#55 · Clinical Chemistry

How does a delay in analysing an arterial blood gas change the results?

Answer

Cells consume oxygen and make carbon dioxide and lactic acid, so pO_2 and pH fall and pCO_2 rises. Analyze within 30 minutes in a plastic syringe at room temperature.

#56 · Clinical Chemistry

Why must air bubbles be expelled from a blood gas syringe?

Answer

Room air lowers pCO_2 and moves pO_2 toward the value of air, giving inaccurate results.

#57 · Clinical Chemistry

What TSH and free T4 pattern indicates primary hypothyroidism?

Answer

High TSH with low free T4. Subclinical disease shows an abnormal TSH with a normal free T4.

#58 · Clinical Chemistry

What is the hook effect in hCG testing?

Answer

A very high hCG saturates the assay antibodies and produces a falsely low or negative result, which is corrected by diluting the sample.

#59 · Clinical Chemistry

From what time after a single acute ingestion is the Rumack-Matthew nomogram valid?

Answer

From 4 hours after ingestion. A result above the treatment line means N-acetylcysteine should be started.

#60 · Clinical Chemistry

What iron study pattern is typical of iron deficiency?

Answer

Low ferritin, low serum iron, high TIBC and low transferrin saturation.

#61 · Hematology

Which anticoagulant is used for the CBC, and how does it prevent clotting?

Answer

EDTA in the lavender-top tube. It binds calcium, which blocks the clotting cascade, and it preserves cell size and shape better than other anticoagulants.

#62 · Hematology

How many times should an EDTA tube be inverted after collection, and why must it be filled to the line?

Answer

Invert gently 8 to 10 times. Underfilling gives excess EDTA, which shrinks red cells and lowers the hematocrit and MCV.

#63 · Hematology

Where is the skin puncture made on an infant, and what is the maximum depth?

Answer

On the lateral or medial plantar surface of the heel, not deeper than 2.0 mm, to avoid striking the heel bone.

#64 · Hematology

What is the formula for MCV, and what is the result with Hct 0.40 L/L and RBC $4.5 \times 10^{12}/L$?

Answer

MCV equals Hct (L/L) x 1000 divided by RBC ($\times 10^{12}/L$). The answer is 88.9 fL.

#65 · Hematology

How are MCH and MCHC calculated in SI units?

Answer

MCH equals hemoglobin (g/L) divided by RBC ($\times 10^{12}/L$), in picograms. MCHC equals hemoglobin (g/L) divided by hematocrit (L/L), in g/L.

#66 · Hematology

What does a raised RDW indicate, and which condition has a low MCV with a normal RDW?

Answer

A raised RDW means increased variation in red cell size (anisocytosis), as in iron deficiency. Thalassemia trait typically has a low MCV with a normal RDW.

#67 · Hematology

How does the Coulter impedance principle count and size a cell?

Answer

A cell passing through an aperture in a conductive fluid raises resistance briefly. The number of pulses is the count, and the pulse height is proportional to cell volume.

#68 · Hematology

What do forward scatter and side scatter measure in flow cytometry hematology analyzers?

Answer

Forward scatter reflects cell size. Side scatter reflects internal complexity such as granules and nuclear lobulation.

#69 · Hematology

What CBC pattern suggests a cold agglutinin, and what is the fix?

Answer

A low RBC count with a raised MCV and raised MCHC. Warm the specimen to 37 degrees Celsius and repeat the analysis.

#70 · Hematology

What is the purpose of a delta check, and what should you do when it fails?

Answer

It compares a result with the patient's previous one to detect wrong blood in tube or mislabelling. Confirm identity, check for clots and repeat before reporting.

#71 · Hematology

How do you correct a WBC count for nucleated red blood cells?

Answer

Corrected WBC equals the analyzer WBC x 100 divided by (100 plus the nRBC per 100 white cells). For example, 12.0 with 20 nRBC gives $10.0 \times 10^9/L$.

#72 · Hematology

What causes EDTA-induced pseudothrombocytopenia, and how is it resolved?

Answer

Antibodies in EDTA clump platelets and lower the count. Recollect in sodium citrate, re-run and multiply the platelet count by 1.1 for the dilution.

#73 · Hematology

What pH range should the Wright-Giemsa buffer have, and what happens if it is too alkaline?

Answer

The buffer pH should be 6.8 to 7.2. Too alkaline makes red cells look too blue and the whole smear dark.

#74 · Hematology

How is the corrected reticulocyte count calculated?

Answer

Reticulocyte percentage x patient hematocrit divided by 0.45. For 6 percent reticulocytes and a hematocrit of 0.30, the corrected count is 4.0 percent.

#75 · Hematology

What are the key conditions for the Westergren ESR?

Answer

A 200 mm tube kept perfectly upright at room temperature, blood diluted 4 to 1 with sodium citrate, and the result read at exactly 60 minutes.

#76 · Hematology

What is the maturation sequence of red cell precursors in the marrow?

Answer

Pronormoblast, basophilic normoblast, polychromatic normoblast, orthochromatic normoblast, reticulocyte, then the mature erythrocyte. The nucleus is extruded at the orthochromatic stage.

#77 · Hematology

What is the Mentzer index, and how does it separate iron deficiency from thalassemia trait?

Answer

MCV divided by RBC count. A value above 13 favours iron deficiency, and below 13 favours thalassemia trait.

#78 · Hematology

Which iron study pattern is typical of iron deficiency anemia?

Answer

Low ferritin, low serum iron, raised TIBC and low transferrin saturation. Ferritin is the best single test because it reflects iron stores.

#79 · Hematology

How do iron studies differ in anemia of chronic disease?

Answer

Serum iron is low and TIBC is low or normal, but ferritin is normal or high because hepcidin traps iron inside macrophages.

#80 · Hematology

What are the smear features of megaloblastic anemia?

Answer

Macro-ovalocytes, marked anisopoikilocytosis and hypersegmented neutrophils, with MCV often above 110 fL, plus a raised LDH and indirect bilirubin from ineffective erythropoiesis.

#81 · Hematology

Which test separates vitamin B12 deficiency from folate deficiency when both raise homocysteine?

Answer

Methylmalonic acid. It is raised in B12 deficiency and normal in folate deficiency.

#82 · Hematology

What laboratory pattern indicates hemolysis?

Answer

Raised reticulocytes, raised LDH and indirect bilirubin, and low haptoglobin, with polychromasia on the smear.

#83 · Hematology

What does the direct antiglobulin test detect, and how does it differ from the indirect test?

Answer

The DAT detects antibody or complement already bound to the patient's red cells. The indirect test detects free antibody in the patient's serum using reagent cells.

#84 · Hematology

Which cell shape and test result separate hereditary spherocytosis from warm autoimmune hemolytic anemia?

Answer

Both show spherocytes. The DAT is negative in hereditary spherocytosis and positive in warm autoimmune hemolytic anemia.

#85 · Hematology

What flow cytometry finding is diagnostic of paroxysmal nocturnal hemoglobinuria?

Answer

Loss of the complement-regulating proteins CD55 and CD59 on the red cells, and the granulocytes and monocytes.

#86 · Hematology

How are Heinz bodies detected, and which condition do they suggest?

Answer

With a supravital stain such as crystal violet or new methylene blue, not Wright stain. They suggest G6PD deficiency or unstable hemoglobin.

#87 · Hematology

Why can G6PD testing be falsely normal after a hemolytic crisis?

Answer

The oldest, most enzyme-deficient cells have been destroyed, and the young reticulocytes have higher enzyme activity. Repeat the test some weeks later.

#88 · Hematology

What does a positive sickle solubility test mean, and what must follow?

Answer

It shows hemoglobin S is present but cannot separate trait from disease. Confirm with electrophoresis or HPLC.

#89 · Hematology

What hemoglobin A2 result is typical of beta thalassemia trait?

Answer

A raised hemoglobin A2, usually about 3.5 to 7 percent, with a mild microcytic anemia and a raised RBC count.

#90 · Hematology

In alkaline hemoglobin electrophoresis, which band travels farthest, and which hemoglobins co-migrate near the origin?

Answer

Hemoglobin A travels farthest toward the anode. Hemoglobins A2, C, E and O stay together near the origin and are separated on acid citrate agar.

#91 · Hematology

What is the order of granulocyte maturation from the earliest recognizable cell to the mature cell in blood?

Answer

Myeloblast, promyelocyte, myelocyte, metamyelocyte, band neutrophil, then segmented neutrophil. The cell shrinks and the nucleus becomes more condensed and lobed as it matures.

#92 · Hematology

Which inclusion is never seen in reactive conditions and confirms a myeloid malignancy when found in blasts?

Answer

The Auer rod, a needle-shaped red-purple inclusion in the cytoplasm. Bundles of Auer rods in abnormal promyelocytes (faggot cells) suggest acute promyelocytic leukemia.

#93 · Hematology

What smear findings suggest a bacterial infection rather than leukemia in a patient with a high neutrophil count?

Answer

Toxic granulation, Dohle bodies, cytoplasmic vacuoles, and a left shift with few or no blasts. This picture is called a leukemoid reaction when the count is very high.

#94 · Hematology

How does the LAP or NAP score help separate a leukemoid reaction from chronic myeloid leukemia?

Answer

The score is high in a leukemoid reaction and low in CML. Molecular detection of BCR::ABL1 is now the definitive test for CML.

#95 · Hematology

Describe the reactive lymphocytes typically seen in infectious mononucleosis.

Answer

They are large and variable in size and shape, with abundant blue cytoplasm that often scallops around neighbouring red cells. Their variation from cell to cell distinguishes them from neoplastic cells.

#96 · Hematology

What blast percentage generally defines acute myeloid leukemia under the current WHO classification?

Answer

Generally 20% or more blasts in the marrow or blood, although certain recurrent genetic abnormalities allow the diagnosis at lower blast counts.

#97 · Hematology

Why must suspected acute promyelocytic leukemia be called immediately by the technologist?

Answer

APL, with t(15;17) and PML::RARA, is strongly associated with disseminated intravascular coagulation. Treatment with all-trans retinoic acid should start as soon as it is suspected.

#98 · Hematology

Which chromosomal abnormality and fusion gene define chronic myeloid leukemia?

Answer

The Philadelphia chromosome, t(9;22), producing the BCR::ABL1 fusion gene. It is monitored by PCR to follow response to tyrosine kinase inhibitor treatment.

#99 · Hematology

What cell is a hallmark of chronic lymphocytic leukemia on the blood smear?

Answer

The smudge cell, also called a basket cell, a fragile lymphocyte ruptured when the smear is made. CLL lymphocytes are small, mature-looking, and usually CD5, CD19, and CD23 positive.

#100 · Hematology

Which cytochemical stain is positive in myeloblasts and negative in lymphoblasts?

Answer

Myeloperoxidase (MPO). Sudan black B behaves similarly. A positive stain supports myeloid lineage, and a negative stain with lymphoblasts supports ALL.

#101 · Hematology

Which stain identifies hairy cell leukemia, and which marker on flow cytometry supports it?

Answer

Tartrate-resistant acid phosphatase (TRAP) is positive in hairy cell leukemia. CD103 expression on flow cytometry also supports the diagnosis.

#102 · Hematology

What three measurements does a flow cytometer record for each cell?

Answer

Forward scatter, which reflects cell size, side scatter, which reflects internal complexity or granularity, and fluorescence from labelled antibodies against markers such as CD34, CD3, or CD19.

#103 · Hematology

What are the typical appearance and blood smear clue of plasma cells in multiple myeloma?

Answer

Plasma cells have an eccentric nucleus, a clock-face chromatin pattern, deep blue cytoplasm, and a pale perinuclear hof. The blood smear often shows rouleaux from high serum protein.

#104 · Hematology

Which anticoagulant is generally used for bone marrow morphology and which for cytogenetics?

Answer

EDTA is generally used for morphology smears, and sodium heparin is generally used for cytogenetics. Always follow the laboratory's own specimen requirements for each test.

#105 · Hematology

Which coagulation pathways does the prothrombin time (PT) measure, and which does the aPTT measure?

Answer

The PT measures the extrinsic and common pathways. The aPTT measures the intrinsic and common pathways. Factor VII deficiency affects only the PT.

#106 · Hematology

What is the required blood to citrate ratio for a coagulation tube, and what minimum fill is acceptable?

Answer

Nine parts blood to one part 3.2% sodium citrate. Tubes must be filled to at least 90% of the stated draw volume, or the sample is rejected.

#107 · Hematology

Why do underfilled citrate tubes give falsely prolonged clotting times?

Answer

Too much citrate is present for the volume of blood, so the sample is diluted and added calcium is bound, making clotting times longer than the patient's true values.

#108 · Hematology

What adjustment is needed for a patient with a hematocrit above 55% when collecting a coagulation sample?

Answer

The citrate volume must be reduced according to the laboratory's approved calculation, because there is less plasma per volume of blood and the citrate is relatively excessive.

#109 · Hematology

Why should coagulation samples not be refrigerated before testing?

Answer

Cold can activate factor VII and platelets, which alters results. Samples are kept capped at room temperature and centrifuged promptly to make platelet-poor plasma.

#110 · Hematology

How is a mixing study performed and what does correction of the prolonged time indicate?

Answer

Patient plasma is mixed 1:1 with normal pooled plasma and retested. Correction into the normal range indicates a factor deficiency, while no correction indicates an inhibitor.

#111 · Hematology

Which inhibitor is the most common cause of a prolonged aPTT that fails to correct on mixing?

Answer

Lupus anticoagulant, an antiphospholipid antibody. Heparin contamination can look similar, so the thrombin time helps rule it out.

#112 · Hematology

Which factor is deficient in hemophilia A and in hemophilia B, and what is the typical PT and aPTT pattern?

Answer

Hemophilia A is factor VIII deficiency and hemophilia B is factor IX deficiency. The aPTT is prolonged and corrects on mixing, and the PT is normal.

#113 · Hematology

What is the most common inherited bleeding disorder, and what does it involve?

Answer

Von Willebrand disease, a deficiency or defect of von Willebrand factor, which carries factor VIII and helps platelets adhere. It causes mostly mucosal bleeding.

#114 · Hematology

What is the typical laboratory pattern in disseminated intravascular coagulation?

Answer

Low platelets, prolonged PT and aPTT, low fibrinogen, markedly raised D-dimer, and often schistocytes on the smear. Trends over time are more useful than one value.

#115 · Hematology

How is unfractionated heparin monitored, and what is the advantage of the anti-Xa assay?

Answer

By the aPTT or anti-Xa assay. Anti-Xa is chromogenic and measures heparin effect directly, so lupus anticoagulant and low factor levels do not interfere.

#116 · Hematology

What is the usual therapeutic INR range for warfarin in most conditions?

Answer

An INR of 2.0 to 3.0 for most conditions. A higher target of 2.5 to 3.5 is used for some mechanical heart valves.

#117 · Hematology

Why is thrombophilia testing for protein C, protein S, and antithrombin often delayed?

Answer

Acute clotting, heparin, and warfarin can lower these proteins and cause falsely abnormal results. Gene tests such as factor V Leiden are not affected by anticoagulants.

#118 · Hematology

Which coagulation analyzer method uses a synthetic substrate that releases colour, and which tests use it?

Answer

The chromogenic method. It is used for anti-Xa, antithrombin, and some protein C assays because it measures enzyme activity directly rather than clot formation.

#119 • Transfusion Medicine

What is the minimum identification required on a pretransfusion sample label in most Canadian hospitals?

Answer

At least two unique patient identifiers, such as full legal name and a health card or medical record number, plus the collection date and collector identification.

#120 • Transfusion Medicine

Why are pre-labelled tubes never acceptable for transfusion testing?

Answer

A label prepared away from the patient can be attached to the wrong tube, so labelling must happen at the bedside with the patient present.

#121 • Transfusion Medicine

How long is a pretransfusion sample usually valid if the patient was transfused or pregnant within the past 3 months?

Answer

Three days, counting the day of collection as day 0, so the sample expires at the end of day 3.

#122 • Transfusion Medicine

How long must a transfusion sample be retained after the transfusion, and at what temperature?

Answer

At least 7 days after transfusion, stored at 1 to 6 degrees Celsius, so testing can be repeated if a reaction is reported.

#123 · Transfusion Medicine

What do the A and B genes actually code for?

Answer

Glycosyltransferase enzymes that add N-acetylgalactosamine (A) or D-galactose (B) to H substance. The O gene produces no active enzyme.

#124 · Transfusion Medicine

What is the Bombay phenotype and how does it type?

Answer

Lack of H, A and B antigens because of no working H gene. Cells type as group O, serum contains anti-H, and only Bombay red cells are compatible.

#125 · Transfusion Medicine

What does the forward group detect, and what does the reverse group detect?

Answer

The forward group tests patient red cells with anti-A and anti-B to detect antigens. The reverse group tests serum with A1 and B cells to detect antibodies.

#126 · Transfusion Medicine

Why are infants younger than about 4 months typed by forward group only?

Answer

Newborns do not make their own ABO antibodies until about 3 to 6 months of age, and antibodies found earlier are usually maternal IgG.

#127 · Transfusion Medicine

What is the difference between weak D and partial D?

Answer

Weak D has fewer but complete D antigens and is usually handled as D positive. Partial D lacks some D epitopes, can form anti-D, and needs D-negative blood.

#128 · Transfusion Medicine

Which test is used to detect weak D expression?

Answer

The indirect antiglobulin test with reagent anti-D, performed on cells that type negative on direct testing.

#129 · Transfusion Medicine

Why are check cells added to every negative antiglobulin test?

Answer

Weakly sensitized check cells must agglutinate. If they do not, the negative result is invalid because of washing failure or neutralized antiglobulin.

#130 · Transfusion Medicine

How often are reagent anti-A, anti-B and anti-D checked with known positive and negative cells?

Answer

On every day of use, and results are recorded. Patient testing stops if any control gives the wrong reaction.

#131 · Transfusion Medicine

How do you separate rouleaux from true agglutination?

Answer

Use a saline replacement technique: remove serum, add saline and re-read. Rouleaux disperses, while true agglutination remains.

#132 · Transfusion Medicine

What are the four groups of ABO discrepancies?

Answer

Group I weak or missing antibodies, group II weak or missing antigens, group III protein or plasma abnormalities such as rouleaux, group IV miscellaneous causes such as cold autoantibodies.

#133 · Transfusion Medicine

Which lectin is used to distinguish A1 cells from A2 cells?

Answer

Dolichos biflorus lectin, which agglutinates A1 cells but not A2 cells.

#134 · Transfusion Medicine

What is the principle of the indirect antiglobulin test (IAT)?

Answer

Antibody coats cells at 37 C, unbound globulin is washed away, and antihuman globulin links coated cells together to give visible agglutination.

#135 · Transfusion Medicine

Why must red cells be washed thoroughly before adding AHG reagent?

Answer

Leftover serum globulin neutralizes the AHG and causes a false negative result, so check cells are added to every negative test.

#136 · Transfusion Medicine

Which enhancement medium must be used only with anti-IgG antiglobulin reagent?

Answer

Polyethylene glycol (PEG), because polyspecific AHG can give false positive reactions with it.

#137 · Transfusion Medicine

What does LISS do in antibody detection?

Answer

Low ionic strength saline increases the rate of antibody uptake onto red cells and shortens incubation to about 10 to 15 minutes.

#138 · Transfusion Medicine

Why are homozygous panel cells preferred for ruling out an antibody?

Answer

They carry a double dose of the antigen, so even a weak antibody showing dosage will react, making a nonreactive result a reliable rule-out.

#139 · Transfusion Medicine

What is the rule of three for identifying an antibody?

Answer

At least three antigen-positive cells react and three antigen-negative cells do not, giving a chance probability of 5 percent or lower.

#140 · Transfusion Medicine

Which blood group antibodies are usually IgG and clinically significant?

Answer

Antibodies in the Rh, Kell, Duffy and Kidd systems, and anti-S and anti-s, react by IAT and can cause hemolytic reactions.

#141 · Transfusion Medicine

Why are Kidd antibodies dangerous even when the screen is negative?

Answer

Anti-Jka and anti-Jkb fall below detection within weeks, so an unrecognized antibody can cause a delayed hemolytic reaction after transfusion.

#142 · Transfusion Medicine

How do ficin and papain treatment change panel reactions?

Answer

Enzymes enhance Rh, Kidd, Lewis, P1 and I reactions, and destroy or weaken Duffy, M and N antigens.

#143 · Transfusion Medicine

When is autoadsorption not appropriate?

Answer

When the patient was transfused within the past 3 months, since donor cells mixed with the patient's cells could absorb an alloantibody.

#144 · Transfusion Medicine

What is the purpose of an elution?

Answer

To free antibody from coated red cells and recover it in a small volume of fluid so the specificity can be identified.

#145 · Transfusion Medicine

What do IgG only and C3d only DAT results usually suggest?

Answer

IgG alone suggests a warm autoantibody or a drug, and C3d alone suggests a cold agglutinin bound with complement.

#146 · Transfusion Medicine

When do cold autoantibodies become clinically significant?

Answer

When they react at 37 C, measured by thermal amplitude. Prewarming or anti-IgG reagent avoids the cold reaction.

#147 · Transfusion Medicine

When may an immediate spin crossmatch be used?

Answer

Only when the antibody screen is negative and the patient has no history of a clinically significant antibody, as it detects ABO incompatibility.

#148 · Transfusion Medicine

What conditions must be met for an electronic crossmatch?

Answer

Two ABO determinations, a negative screen with no significant antibody history, and a validated computer system that blocks release of incompatible units.

#149 · Transfusion Medicine

What is the required storage temperature range for red blood cell units in a blood bank refrigerator?

Answer

Red cells are stored at 1 to 6 degrees Celsius. The temperature is recorded continuously and the refrigerator has an audible alarm for out-of-range readings.

#150 · Transfusion Medicine

At what temperature and under what condition are platelet components stored?

Answer

Platelets are stored at 20 to 24 degrees Celsius with constant gentle agitation. They are never refrigerated because cold storage shortens their survival after transfusion.

#151 · Transfusion Medicine

Why do platelets carry the highest risk of bacterial contamination among the components?

Answer

They are stored at room temperature, which allows bacteria to grow. Fever, rigors and shock during or after a platelet transfusion should raise suspicion of sepsis.

#152 · Transfusion Medicine

Which coagulation factors and proteins does cryoprecipitate concentrate?

Answer

Cryoprecipitate concentrates fibrinogen, factor VIII, von Willebrand factor and factor XIII. Its main use is fibrinogen replacement in a bleeding patient.

#153 · Transfusion Medicine

What is the purpose of irradiating cellular blood components?

Answer

Irradiation stops donor lymphocytes from dividing and prevents transfusion-associated graft-versus-host disease. The minimum dose is 25 Gy to the centre of the unit.

#154 · Transfusion Medicine

Can leukoreduction replace irradiation for preventing TA-GVHD?

Answer

No. Enough lymphocytes can remain in a leukoreduced unit to cause TA-GVHD, so patients who require irradiated components still need irradiation.

#155 · Transfusion Medicine

How does irradiation change the shelf life of red cells, and why?

Answer

Shelf life is shortened to 28 days after irradiation or the original outdate, whichever comes first, because irradiated red cells leak potassium faster during storage.

#156 · Transfusion Medicine

What are the benefits of prestorage leukoreduction of blood components?

Answer

It reduces febrile non-hemolytic reactions, lowers HLA alloimmunization and reduces CMV transmission, since CMV is carried in white cells.

#157 • Transfusion Medicine

What is the standard time limit for starting and completing a red cell transfusion after issue?

Answer

Start within 30 minutes of leaving controlled storage and complete within four hours of the start. Otherwise the unit is returned or discarded per policy.

#158 • Transfusion Medicine

What is the usual cause of an acute hemolytic transfusion reaction?

Answer

ABO-incompatible red cells given because of a clerical or patient identification error. Complement-mediated destruction of the cells causes fever, hemoglobinuria, low blood pressure and DIC.

#159 · Transfusion Medicine

What are the first steps at the bedside when a transfusion reaction is suspected?

Answer

Stop the transfusion, keep the intravenous line open with saline, recheck patient and unit identification, and notify the physician and the blood bank immediately.

#160 · Transfusion Medicine

How do TRALI and TACO differ in cause and response to diuretics?

Answer

TRALI is noncardiogenic, caused by donor antibodies, with normal filling pressure and no diuretic benefit. TACO is circulatory overload with raised BNP that improves with diuretics.

#161 • Transfusion Medicine

What laboratory checks make up the initial transfusion reaction workup?

Answer

A clerical check, visual inspection of pre and post plasma for hemolysis, a repeat ABO and Rh on the post sample, and a direct antiglobulin test on the post sample.

#162 • Transfusion Medicine

When do delayed hemolytic transfusion reactions occur and what laboratory finding is typical?

Answer

They occur three to 14 days after transfusion in previously sensitized patients. Typical findings are a falling hemoglobin, a positive DAT and a new antibody, often Kidd or Rh.

#163 · Transfusion Medicine

Which immunoglobulin class causes HDFN and why?

Answer

IgG causes HDFN because it is the only antibody class that crosses the placenta. It coats fetal red cells carrying the matching antigen and they are removed by the fetal spleen.

#164 · Transfusion Medicine

Which maternal-baby ABO combination most often causes ABO HDFN, and how severe is it usually?

Answer

A group O mother with a group A or B baby is the classic combination. The disease is usually mild, and the baby's DAT is often weakly positive or negative.

#165 · Transfusion Medicine

Why is anti-Kell HDFN harder to predict from the antibody titre than anti-D?

Answer

Anti-Kell can suppress the production of fetal red cell precursors in the marrow as well as destroy mature cells, so the titre does not reflect severity as well.

#166 · Transfusion Medicine

How is an antibody titre reported and what change in titre is considered significant?

Answer

The titre is the reciprocal of the highest dilution giving a 1+ reaction. A rise of two dilutions or more, tested in parallel with the previous sample, is significant.

#167 • Transfusion Medicine

What schedule of Rh immune globulin is standard for an RhD-negative pregnant woman without anti-D?

Answer

One 300 microgram dose at about 28 weeks, and another within 72 hours after delivery of an RhD-positive baby. Extra doses follow sensitizing events such as miscarriage or trauma.

#168 • Transfusion Medicine

How much fetal whole blood does one 300 microgram dose of RhIG cover?

Answer

One 300 microgram dose covers up to 30 mL of fetal whole blood, which is about 15 mL of fetal red cells.

#169 · Transfusion Medicine

What does the rosette test detect and how does it work?

Answer

It is a qualitative screen for fetal D-positive cells in maternal blood. Anti-D coats them, and indicator D-positive cells cling to them, forming rosettes seen microscopically.

#170 · Transfusion Medicine

In the Kleihauer-Betke test, which cells stain pink and why?

Answer

Fetal cells stain pink because fetal hemoglobin resists acid elution. Adult hemoglobin dissolves out in the acid buffer, leaving pale ghost cells.

#171 • Transfusion Medicine

Calculate the RhIG vials needed if a Kleihauer-Betke count shows 0.6 percent fetal cells.

Answer

Fetal bleed is $0.006 \times 5,000 \text{ mL} = 30 \text{ mL}$, which is 1 vial. Adding one extra vial for error gives a total of 2 vials.

#172 • Transfusion Medicine

What attributes are required for red cells used in intrauterine transfusion?

Answer

Group O, antigen-negative for the maternal antibody, irradiated, CMV-safe, fresh, concentrated, and crossmatch-compatible with the maternal plasma.

#173 · Transfusion Medicine

Why is reverse ABO grouping not required in infants under four months of age?

Answer

Infants do not make their own anti-A or anti-B at this age. Any ABO antibodies found are maternal in origin and do not reflect the infant's true group.

#174 · Transfusion Medicine

What ratio of components does a modern massive transfusion protocol aim for?

Answer

A balanced ratio of red cells, plasma and platelets close to one to one to one. The protocol is released as a pre-planned pack for a bleeding patient.

#175 · Transfusion Medicine

What special blood products does an IgA-deficient patient with anti-IgA require?

Answer

Washed red cells and platelets or products from IgA-deficient donors, and plasma only from IgA-deficient donors, to avoid severe anaphylactic reactions.

#176 · Transfusion Medicine

How is the corrected count increment calculated and what does a poor result suggest?

Answer

CCI equals the count rise in $\times 10^9/L$ times body surface area in square metres divided by platelets given in units of 10^{11} . A poor result on repeated tests suggests immune refractoriness.

#177 · Clinical Microbiology

Why is CSF for bacterial culture never refrigerated, and how is it transported to the laboratory?

Answer

Neisseria meningitidis and *Haemophilus influenzae* are fragile and die in the cold. CSF is delivered immediately at room temperature and processed as a stat specimen.

#178 · Clinical Microbiology

What is the recommended blood volume per bottle for an adult blood culture, and why does volume matter?

Answer

About 8 to 10 mL per bottle. Bacteremia often involves very few organisms per mL, so blood volume is the strongest driver of recovery.

#179 · Clinical Microbiology

Within what time should a routine urine specimen reach the laboratory if it is not preserved?

Answer

Within two hours, or it should be refrigerated or collected in a boric acid preservative tube, since bacteria multiply rapidly in urine at room temperature.

#180 · Clinical Microbiology

What squamous epithelial cell count per low-power field usually leads to rejection of a sputum specimen?

Answer

More than 25 squamous epithelial cells per low-power field. This shows the sample is mostly saliva contaminated with oral flora.

#181 · Clinical Microbiology

What is the purpose of Gram iodine in the Gram stain procedure?

Answer

It acts as a mordant, forming a large crystal violet-iodine complex inside the cell that gram-positive walls can trap during decolourization.

#182 · Clinical Microbiology

Which step of the Gram stain is the most common source of error, and what happens if it is overdone?

Answer

Decolourization. Too much decolourizer makes gram-positive cells look pink and gram-negative, while too little leaves gram-negative cells purple.

#183 · Clinical Microbiology

Why does chocolate agar support *Haemophilus influenzae* when sheep blood agar does not?

Answer

Heating lyses the red cells and releases hemin (X factor) and NAD (V factor), both of which *Haemophilus* requires for growth.

#184 · Clinical Microbiology

What do lactose fermenters look like on MacConkey agar, and why?

Answer

They form pink colonies because acid from lactose fermentation changes the neutral red indicator. Nonfermenters stay colourless.

#185 · Clinical Microbiology

How is a urine colony count calculated using a 0.001 mL calibrated loop?

Answer

Multiply the colonies counted by 1000 to get CFU/mL. For example, 45 colonies equals 45 000 CFU/mL.

#186 · Clinical Microbiology

What atmosphere and temperature does *Campylobacter* need for culture?

Answer

A microaerophilic atmosphere of about 5 percent oxygen and 10 percent carbon dioxide, incubated at 42 degrees Celsius.

#187 · Clinical Microbiology

Which simple test separates Staphylococcus from Streptococcus, and what result does each give?

Answer

The catalase test. Staphylococcus is catalase positive and produces bubbles with hydrogen peroxide, while Streptococcus is catalase negative.

#188 · Clinical Microbiology

Why should catalase testing not be done on colonies taken from blood agar?

Answer

Red blood cells contain catalase, so carried-over blood can produce bubbles and a false positive result.

#189 · Clinical Microbiology

Which reagent is added in the indole test, and what colour indicates a positive result?

Answer

Kovac reagent is added. A red layer at the top of the medium shows indole was produced by tryptophanase, as with *Escherichia coli*.

#190 · Clinical Microbiology

Which containment level applies to most routine diagnostic bacteriology, and what cabinet work does it require?

Answer

Containment level 2. Procedures that may create aerosols are done inside a biological safety cabinet, with gloves, lab coat, and restricted access.

#191 · Clinical Microbiology

What three checks make up quality control of a new lot of culture medium?

Answer

Sterility testing by incubating sample plates, physical inspection for defects, and performance testing with known reference organisms giving expected growth and reactions.

#192 · Clinical Microbiology

Which test separates *Staphylococcus aureus* from coagulase-negative staphylococci, and what is the result for *S. aureus*?

Answer

The coagulase test. *Staphylococcus aureus* is coagulase positive, while *Staphylococcus epidermidis* and other coagulase-negative species do not clot plasma.

#193 · Clinical Microbiology

Which coagulase-negative *Staphylococcus* is novobiocin resistant and causes urinary tract infection in young women?

Answer

Staphylococcus saprophyticus. It is novobiocin resistant, which separates it from *Staphylococcus epidermidis*, which is susceptible.

#194 · Clinical Microbiology

How do group A and group B streptococci differ in key laboratory tests?

Answer

Group A (*S. pyogenes*) is PYR positive and bacitracin susceptible. Group B (*S. agalactiae*) is hippurate positive and CAMP positive.

#195 · Clinical Microbiology

Which tests identify *Streptococcus pneumoniae* and separate it from viridans streptococci?

Answer

S. pneumoniae is optochin susceptible and bile soluble. Viridans streptococci are optochin resistant and bile insoluble.

#196 · Clinical Microbiology

Which results identify *Enterococcus*?

Answer

Enterococcus is catalase negative, PYR positive, bile esculin positive, and grows in 6.5 percent sodium chloride.

#197 · Clinical Microbiology

At what gestational age are pregnant women screened for group B Streptococcus, and with what specimen?

Answer

At 35 to 37 weeks, using a combined vaginal and rectal swab, to guide prevention of neonatal infection.

#198 · Clinical Microbiology

What are the features of *Listeria monocytogenes* in the laboratory and its clinical significance?

Answer

A small gram-positive rod with narrow beta hemolysis and tumbling motility that grows in the cold. It causes meningitis in newborns, pregnant women, and immunocompromised patients.

#199 · Clinical Microbiology

What specimen is appropriate for *Clostridioides difficile* testing?

Answer

Liquid or unformed stool that takes the shape of the container. Formed stool is rejected, since it does not reflect active disease.

#200 · Clinical Microbiology

What is the first branching step for identifying Enterobacterales on MacConkey agar?

Answer

Lactose fermentation. Lactose fermenters such as *Escherichia coli* and *Klebsiella* form pink colonies, while *Salmonella*, *Shigella*, and *Proteus* stay colourless.

#201 · Clinical Microbiology

How do *Escherichia coli* and *Klebsiella pneumoniae* differ in indole and urease reactions?

Answer

Escherichia coli is indole positive and urease negative. *Klebsiella pneumoniae* is indole negative and urease positive, and it forms mucoid colonies.

#202 · Clinical Microbiology

Which features identify *Proteus mirabilis* in culture?

Answer

Swarming on blood agar, lactose negative on MacConkey, strongly urease positive, and hydrogen sulfide positive.

#203 · Clinical Microbiology

What identifies *Pseudomonas aeruginosa* among gram-negative rods?

Answer

It is oxidase positive, grows at 42 degrees Celsius, produces blue-green pigment, and has a grape-like odour. Enterobacterales are oxidase negative.

#204 · Clinical Microbiology

Why does *Haemophilus influenzae* grow on chocolate agar but not on plain sheep blood agar?

Answer

It needs X factor (hemin) and V factor (NAD). Chocolate agar supplies both, while sheep blood agar lacks free V factor, except near a *Staphylococcus* streak.

#205 · Clinical Microbiology

What growth conditions and colony colour identify *Vibrio cholerae* and *Campylobacter jejuni*?

Answer

Vibrio cholerae grows as yellow sucrose-fermenting colonies on TCBS agar. *Campylobacter jejuni* needs microaerophilic conditions at 42 degrees Celsius.

#206 · Clinical Microbiology

Why do mycobacteria stain acid-fast and what colour do they appear with the Ziehl-Neelsen method?

Answer

Their mycolic acid-rich waxy wall resists acid-alcohol decolourization, so they retain carbol fuchsin and appear red against a blue or green background.

#207 · Clinical Microbiology

What McFarland turbidity standard is used to adjust the inoculum for disk diffusion testing, and roughly what cell density does it represent?

Answer

A 0.5 McFarland standard is used. It corresponds to roughly 1.5×10^8 colony-forming units per millilitre of bacterial suspension.

#208 · Clinical Microbiology

What is the required depth of Mueller-Hinton agar for disk diffusion, and what happens if the agar is too thin?

Answer

The agar depth should be 4 mm. Agar that is too thin gives larger zones of inhibition and can cause false susceptibility.

#209 · Clinical Microbiology

How does a too-heavy inoculum affect Kirby-Bauer zone diameters and the reported result?

Answer

A too-heavy inoculum shrinks the zones of inhibition, which can make a susceptible organism appear resistant. This is a false resistant result.

#210 · Clinical Microbiology

What is the definition of the minimum inhibitory concentration (MIC)?

Answer

The MIC is the lowest concentration of an antibiotic that prevents visible growth of an organism after overnight incubation. It is reported in mg/L.

#211 · Clinical Microbiology

Which MIC method is considered the reference method, and what final inoculum does it use?

Answer

Broth microdilution is the reference method. The final inoculum in each well is about 5×10^5 CFU/mL in cation-adjusted Mueller-Hinton broth.

#212 · Clinical Microbiology

How is the MIC read on a gradient strip test when it falls between two printed values?

Answer

The MIC is read where the edge of the elliptical zone meets the strip. If it falls between marks, the higher marked value is reported.

#213 · Clinical Microbiology

Which surrogate disk is used to detect mecA-mediated methicillin resistance in *Staphylococcus aureus*, and what zone indicates resistance?

Answer

The cefoxitin disk is used. A zone diameter of 21 mm or less for *S. aureus* indicates methicillin resistance.

#214 · Clinical Microbiology

What change does the vanA gene cluster make that causes vancomycin resistance in enterococci?

Answer

It changes the cell wall precursor target from D-alanyl-D-alanine to D-alanyl-D-lactate, which sharply reduces vancomycin binding. VanA gives resistance to vancomycin and teicoplanin.

#215 · Clinical Microbiology

How is an ESBL confirmed phenotypically in *Escherichia coli* or *Klebsiella pneumoniae*?

Answer

A cephalosporin disk is compared with the same drug plus clavulanate. An increase of 5 mm or more in the combination zone confirms an ESBL.

#216 · Clinical Microbiology

Why can resistance emerge during therapy with a third-generation cephalosporin in *Enterobacter cloacae*?

Answer

Enterobacter carries an inducible chromosomal AmpC beta-lactamase. Exposure to the drug can select for derepressed mutants, so an initially susceptible isolate becomes resistant during treatment.

#217 · Clinical Microbiology

What does a D-shaped zone around a clindamycin disk placed near an erythromycin disk indicate?

Answer

It is a positive D-test, showing inducible clindamycin resistance from an *erm* gene. Clindamycin should be reported as resistant or with a comment about resistance during therapy.

#218 · Clinical Microbiology

What does the modified carbapenem inactivation method (mCIM) detect, and what does adding EDTA (eCIM) help distinguish?

Answer

The mCIM detects carbapenemase production by an isolate. Adding EDTA helps distinguish metallo-beta-lactamases, which EDTA inhibits, from serine carbapenemases.

#219 · Clinical Microbiology

Which ATCC quality control strain is commonly used for disk diffusion testing of Enterobacterales, and what must happen if it is out of range?

Answer

Escherichia coli ATCC 25922 is used. If a result is out of range, patient results for that drug are held until the cause is found and corrected.

#220 · Clinical Microbiology

Which isolates are counted when a hospital compiles a cumulative antibiogram?

Answer

Only the first isolate of each species per patient per analysis period is counted. This stops a single patient with repeat cultures from skewing the data.

#221 · Clinical Microbiology

Why must a germ tube test be read within 3 hours of incubation in serum?

Answer

Later reading allows other *Candida* species to form pseudo-germ tubes, which can cause false positives. A true germ tube has no constriction at its origin.

#222 · Clinical Microbiology

Which two *Candida* species are germ tube positive?

Answer

Candida albicans and *Candida dubliniensis* are germ tube positive. Species such as *Candida glabrata* are negative and need assimilation panels, chromogenic agar or MALDI-TOF.

#223 · Clinical Microbiology

Which features help identify *Cryptococcus neoformans* in culture?

Answer

It is urease positive, has a polysaccharide capsule, and produces brown colonies on niger seed agar because it makes melanin from phenolic compounds.

#224 · Clinical Microbiology

Why must suspected dimorphic fungi be handled only in a biological safety cabinet?

Answer

Their mold phase makes airborne spores that are easily inhaled, and laboratory-acquired infections are well documented. *Coccidioides* arthroconidia are especially infectious.

#225 · Clinical Microbiology

What does dimorphic mean for a fungus such as *Histoplasma capsulatum*?

Answer

It grows as a mold at 25 to 30 degrees C and as a yeast or spherule in tissue or at 35 to 37 degrees C.

#226 · Clinical Microbiology

How does a KOH preparation help detect fungi in skin, hair or nails?

Answer

The 10 to 20 percent KOH dissolves keratin and cellular debris while leaving fungal cell walls intact, so hyphae or yeast cells become visible.

#227 · Clinical Microbiology

How does calcofluor white reveal fungi under the microscope?

Answer

It binds chitin and cellulose in fungal cell walls and fluoresces bright apple-green or blue-white under ultraviolet light. It is more sensitive than KOH alone.

#228 · Clinical Microbiology

What does an India ink preparation of cerebrospinal fluid show for Cryptococcus?

Answer

Ink particles cannot enter the capsule, so the yeast appears as a round budding cell with a clear halo on a dark background. A negative result does not exclude infection.

#229 · Clinical Microbiology

Why is cryptococcal antigen testing preferred over India ink for detection?

Answer

It detects capsular polysaccharide in CSF or serum and is much more sensitive than India ink. Diluting the specimen resolves a false negative from the prozone effect.

#230 · Clinical Microbiology

What is the purpose of the formalin-ethyl acetate concentration technique in an ova and parasite exam?

Answer

It concentrates parasites into the sediment while fat and debris collect in the top plug. This raises the chance of finding low numbers of organisms.

#231 · Clinical Microbiology

Which stain is used on PVA-preserved stool to identify intestinal protozoa such as Entamoeba?

Answer

Trichrome stain is used. It shows nuclei and internal structures so nuclear count and chromatin pattern can be assessed.

#232 · Clinical Microbiology

Why are both thick and thin Giemsa films needed for suspected malaria?

Answer

The thick film concentrates blood and is more sensitive for finding parasites. The thin film shows red cell morphology, allowing species identification and calculation of parasitemia.

#233 · Clinical Microbiology

Which finding distinguishes Babesia from Plasmodium falciparum on a blood film?

Answer

Babesia can show Maltese cross tetrad forms and has no hemozoin pigment, gametocytes or Schuffner's dots. Its rings can otherwise look like *P. falciparum*.

#234 · Clinical Microbiology

Which swab types should not be used for viral PCR specimens, and why?

Answer

Calcium alginate swabs and wooden shafts should not be used because they can inhibit PCR. Synthetic-tip swabs in viral transport medium are preferred.

#235 · Histology

What is the purpose of accessioning a histology specimen in the laboratory?

Answer

Accessioning logs the specimen into the system and assigns a unique accession number that follows the cassette, block, slides and report, preventing patient mix-ups.

#236 · Histology

What should the technologist do when a specimen label and its requisition do not match?

Answer

Hold the specimen and do not process it until the discrepancy is resolved with the sender, then document the event through the nonconformance process.

#237 · Histology

How does formaldehyde fix tissue at the molecular level?

Answer

It forms cross-links between proteins, mainly between amino groups on neighbouring molecules. These cross-links form slowly, so fixation takes hours even though penetration is quick.

#238 · Histology

What fixative to tissue volume ratio is commonly taught for good fixation?

Answer

The fixative volume should be at least 10 times the volume of the tissue. Some references recommend 15 to 20 times for best results.

#239 · Histology

Why is formalin buffered to a neutral pH, and what happens if it is not?

Answer

Unbuffered formalin becomes formic acid, which reacts with hemoglobin and leaves dark brown-black formalin pigment (acid formaldehyde hematin) in bloody tissue.

#240 · Histology

Which fixative is known for excellent nuclear detail but stains tissue yellow?

Answer

Bouin fluid, which contains picric acid. It is often used for testicular biopsies and must be washed out of the tissue.

#241 · Histology

What are the effects of under-fixation on a processed tissue block?

Answer

The centre stays soft and processes poorly, so the section shows a pale, poorly stained centre with a better stained rim, sometimes called a halo effect.

#242 · Histology

What maximum thickness should tissue pieces have when placed in a cassette?

Answer

Usually no more than 3 to 4 millimetres, so that processing fluids can reach the centre of each piece during the processor run.

#243 · Histology

Why must decalcification be performed after fixation and not before?

Answer

Decalcifying unfixed tissue damages cell detail. The tissue must be fixed first so that the acid or chelator does not destroy its structure.

#244 · Histology

Which decalcifier is gentlest on nucleic acids and antigens but slowest to work?

Answer

EDTA, a chelating agent that binds calcium ions instead of dissolving them. It can take days to weeks for dense bone.

#245 · Histology

How is the chemical end point of an acid decalcifier tested?

Answer

Ammonium hydroxide and ammonium oxalate are added to the used fluid. A cloudy calcium oxalate precipitate means calcium is still being released; repeated clear tests mean it is complete.

#246 · Histology

List the routine tissue processor steps in the correct order.

Answer

Formalin fixation, dehydration in graded alcohols, clearing with xylene or a substitute, then infiltration with molten paraffin wax under heat and usually vacuum.

#247 · Histology

Why does the processor use graded alcohols rather than absolute alcohol first?

Answer

Starting weaker and rising gradually prevents the sudden shrinkage and distortion that would result from placing wet tissue directly into strong alcohol.

#248 · Histology

What does a soft, crumbly block centre after processing usually indicate?

Answer

Incomplete dehydration or incomplete wax infiltration, caused by tired alcohols, pieces that are too thick, or a run that was too short.

#249 · Histology

What are the three main components of WHMIS in Canada?

Answer

Container labels, safety data sheets, and worker education and training. Decanted containers need a workplace label naming the contents and hazards.

#250 · Histology

Why is skin normally embedded on edge rather than flat?

Answer

Embedding on edge lets a single section show all layers, epidermis, dermis and subcutaneous fat, together in one cut.

#251 · Histology

What typical section thickness is used for routine paraffin sections?

Answer

About 3 to 5 micrometres. Thinner sections give better nuclear detail, and thicker sections can look dark and blurry with overlapping cells.

#252 · Histology

What causes chatter on a paraffin section, and how is it corrected?

Answer

A loose blade or block, a dull or wrongly angled blade, or hard tissue. Tighten the clamps, change the blade, adjust the angle, or soften the block with ice.

#253 · Histology

What does a clearance angle that is too small do to sections?

Answer

The block face rubs against the back of the blade, so the sections are compressed. Too large an angle causes scraping or curling.

#254 · Histology

What temperature range is the flotation bath usually kept at, and why?

Answer

About 42 to 48 degrees Celsius, several degrees below the wax melting point. The warmth flattens the section without melting it.

#255 · Histology

Why must the flotation bath surface be skimmed between patient cases?

Answer

To remove tissue fragments that could float onto another patient's slide as a contaminant, also called a floater, causing a diagnostic error.

#256 · Histology

Which slides help fatty, decalcified and immunohistochemistry sections stay attached during staining?

Answer

Positively charged slides or slides coated with an adhesive such as silane or poly-L-lysine. These bind the negatively charged tissue.

#257 · Histology

What causes scoring lines running through every section from a block?

Answer

A nick in the blade or a hard particle such as calcium or suture in the tissue. Move to a fresh blade area or decalcify.

#258 · Histology

Why must hematoxylin be oxidized and combined with a mordant before it stains nuclei?

Answer

Hematoxylin is oxidized to hematein, then bound to a metal mordant such as alum. This positively charged complex binds negatively charged nuclear nucleic acids.

#259 · Histology

What is the difference between progressive and regressive hematoxylin staining?

Answer

Progressive staining stops at the desired intensity. Regressive staining overstains, then removes excess dye with dilute acid alcohol in a differentiation step.

#260 · Histology

What is the purpose of the bluing step in H&E staining?

Answer

Bluing makes the section mildly alkaline so hematoxylin changes from reddish-purple to crisp blue. Weak bluing leaves nuclei reddish and prone to fading.

#261 · Histology

What colours does eosin give, and to which structures does it bind?

Answer

Shades of pink and red. Eosin is an acidic dye that binds positively charged proteins in the cytoplasm, collagen and other connective tissue.

#262 · Histology

What does a cloudy, milky appearance after coverslipping usually mean?

Answer

Water remained in the section before clearing, usually because the final dehydrating alcohols were exhausted or the section was not dehydrated long enough.

#263 · Histology

Why is a cryostat used during surgery?

Answer

It cuts fresh frozen tissue quickly so a pathologist can give a rapid preliminary answer, such as whether a margin is clear or a lesion is malignant.

#264 · Histology

Which tissue must not be cut in the cryostat because of infection risk?

Answer

Tissue from patients known or suspected to have Creutzfeldt-Jakob disease. Fresh tissue also creates aerosols, so standard precautions apply.

#265 · Histology

What does a periodic acid-Schiff (PAS) stain colour, and what colour does it produce?

Answer

PAS stains glycogen, neutral mucins, basement membranes, and fungal cell walls magenta. Periodic acid creates aldehydes from sugars, and Schiff reagent reacts with them.

#266 · Histology

How does a PAS with diastase separate glycogen from other PAS-positive material?

Answer

Diastase digests glycogen. Material that stains without diastase but disappears with it is glycogen, while material still magenta afterward is neutral mucin, fungal wall, or basement membrane.

#267 · Histology

Which stain is the sensitive screening method for fungi and Pneumocystis, and what does it show?

Answer

Gomori methenamine silver (GMS) stains fungi and Pneumocystis black against a pale green background. Silver is reduced onto the oxidized polysaccharide of the organism wall.

#268 · Histology

Why must acid-washed glassware and non-metal forceps be used in silver stains?

Answer

Silver methods are very sensitive to contamination. Dirty glassware or metal contact causes precipitate and background deposits that can mimic organisms or hide detail.

#269 · Histology

What does the Ziehl-Neelsen stain detect, and how do the organisms appear?

Answer

It detects acid-fast bacilli such as *Mycobacterium tuberculosis*. Organisms appear as bright red rods on a blue methylene blue background after acid alcohol decolourization.

#270 · Histology

What confirms that a Congo red stained deposit is amyloid?

Answer

Apple-green birefringence under polarized light confirms amyloid. Congo red gives a pink to red colour, but that alone is not diagnostic without polarization.

#271 · Histology

What does Alcian blue at pH 2.5 stain, and what does Alcian blue with PAS show?

Answer

Alcian blue at pH 2.5 stains acidic mucins blue. Combined with PAS, acidic mucins are blue, neutral mucins are magenta, and mixed mucins appear purple.

#272 · Histology

What is the target and result of the Prussian blue (Perls) stain?

Answer

It detects ferric iron in hemosiderin, which stains bright blue with nuclei counterstained red. Iron-free glassware and water prevent false positive results.

#273 · Histology

What colours do Masson trichrome and Verhoeff-Van Gieson show for collagen and other components?

Answer

Trichrome colours collagen blue or green and muscle red. Verhoeff-Van Gieson colours elastic fibres black, collagen red, and other tissue yellow.

#274 · Histology

What does a reticulin silver stain show, and where is it commonly used?

Answer

It stains thin type III collagen fibres black. It is used to assess the liver plate framework, lymphoid tissue, and bone marrow fibrosis.

#275 · Histology

What is the purpose of a positive control in special stains and immunohistochemistry?

Answer

A positive control is tissue known to contain the target, stained with the patient slide. If it fails, the entire run is invalid and patient results cannot be released.

#276 · Histology

Why is antigen retrieval needed before immunohistochemistry on formalin-fixed tissue?

Answer

Formalin cross-links proteins and can mask epitopes. Antigen retrieval, most often heat-induced in a buffer, reverses the masking so the antibody can bind.

#277 · Histology

What is the advantage of polymer-based detection over avidin-biotin detection in IHC?

Answer

Polymer systems give high sensitivity without biotin, so they avoid false positive background in biotin-rich tissues such as liver and kidney.

#278 · Histology

How is HER2 scored by immunohistochemistry, and what happens to an equivocal result?

Answer

HER2 is a membrane stain scored 0 to 3+. An equivocal score of 2+ is usually followed by in situ hybridization to check for gene amplification.

#279 · Histology

What is the main purpose of an intraoperative frozen section consultation?

Answer

It gives the surgeon a rapid answer, such as margin status or malignancy, while the patient is in surgery. The final diagnosis is made on permanent paraffin sections.

#280 · Histology

What temperature range is a cryostat chamber usually set to for routine frozen sections?

Answer

About minus 20 to minus 25 degrees Celsius. Fatty tissue needs colder settings, around minus 30 degrees Celsius or lower, to cut without smearing.

#281 · Histology

What causes ice-crystal artifact in frozen sections, and how is it reduced?

Answer

Slow freezing lets water form large ice crystals that leave holes in the tissue. Faster freezing produces smaller crystals and better morphology.

#282 · Histology

What does a deferral mean on a frozen section report?

Answer

The pathologist cannot make a safe diagnosis on the frozen section and will wait for permanent sections. It is an acceptable and often safest result.

#283 · Histology

What communication steps are required when a frozen section result is given to the surgeon?

Answer

The pathologist telephones the result, the surgeon reads it back to confirm, and the time and content of the call are documented.

#284 · Histology

Why are muscle biopsies snap frozen in isopentane cooled by liquid nitrogen?

Answer

Many diagnostic enzyme stains do not work after formalin fixation, and cooled isopentane freezes fast enough to prevent ice-crystal damage to muscle fibres.

#285 · Histology

How is a renal biopsy usually divided for testing?

Answer

One portion goes to formalin for light microscopy, one is frozen fresh for immunofluorescence, and one goes into glutaraldehyde for electron microscopy.

#286 · Histology

What is the standard fixative for electron microscopy, and why must the tissue be tiny?

Answer

Buffered glutaraldehyde, often 2 to 3 percent. Pieces are about 1 mm because fixative penetrates slowly and delayed fixation degrades fine structure.

#287 · Histology

What special step is taken in Mohs surgery to allow tumour location on the patient?

Answer

The specimen is oriented and marked with dyes or a map so any positive margin area can be located exactly, and the entire margin is cut in one plane.

#288 · Histology

Why is frozen sectioning of suspected prion disease specimens avoided?

Answer

Prions resist routine fixation and decontamination, so cutting them exposes staff and contaminates equipment. Special protocols agreed with infection control are used instead.

#289 · Histology

What is the greatest immediate injury hazard when working with a cryostat or microtome?

Answer

The blades are extremely sharp, so sharps injuries are the main hazard. Blades are handled with a holder or dispenser and discarded into a sharps container.

#290 · Histology

How is the temperature of a laboratory reagent refrigerator monitored?

Answer

Temperature is logged daily and must stay within about 2 to 8 degrees Celsius. A reading outside the range triggers documented corrective action.

#291 · Histology

What should be done with an instrument that fails a calibration check?

Answer

It is taken out of service and labelled until repaired and confirmed working. The failure and the corrective action are documented.

#292 · Histology

What is a near miss, and why is it reported?

Answer

A near miss is an error caught before it caused harm. Reporting it reveals weak points in a process so the cause can be corrected before real harm occurs.

#293 · Cytology

Which fixation is required for a smear that will be stained with the Papanicolaou stain, and when must it be applied?

Answer

Wet fixation in 95% ethanol, applied immediately after the smear is made and before the cells can dry. Air drying before fixation ruins nuclear detail.

#294 · Cytology

What appearance does air-drying artefact give to cells on a smear meant for the Papanicolaou stain?

Answer

Nuclei look enlarged, pale, and smudged, and cells look flattened. This can cause a false suspicion of abnormality or an unsatisfactory result.

#295 · Cytology

How far from the slide should a spray fixative be held, and what must be done to the coating before staining?

Answer

About 20 to 25 cm from the slide. The protective coating must be removed by soaking in alcohol before the Papanicolaou stain is applied.

#296 · Cytology

How does the ThinPrep system transfer cells onto the slide?

Answer

It uses a membrane-based process. A rotating filter draws cells from the suspension, and a thin layer is pressed onto the slide as a circular spot about 20 mm across.

#297 · Cytology

How does the SurePath system prepare a liquid-based slide?

Answer

It uses a density gradient. Centrifugation removes much of the blood and debris, and cells settle by gravity onto the slide as a round spot of about 13 mm.

#298 · Cytology

Name two advantages of liquid-based cytology over a conventional cervical smear.

Answer

Cells are spread in a thin, even layer with less blood and mucus, and the leftover fluid can be used for extra tests such as HPV testing.

#299 · Cytology

What is a cytospin and which specimens is it commonly used for?

Answer

A cytocentrifuge that flings cells from a small fluid volume onto a slide as an even round spot. It is commonly used for cerebrospinal fluid and other body fluids.

#300 · Cytology

What is a cell block and why is it useful in cytology?

Answer

A pellet of cells held in agar or gel, formalin-fixed, and processed like tissue. It preserves cell groups and permits H&E, immunohistochemistry, and molecular tests.

#301 · Cytology

Which stain uses air-dried smears and is used for rapid bedside assessment of fine needle aspirates?

Answer

Diff-Quik, a rapid Romanowsky-type stain. It shows cytoplasmic detail, mucin, and background material well but gives less crisp nuclear detail than the Papanicolaou stain.

#302 · Cytology

Which dyes colour the nuclei and the cytoplasm in the Papanicolaou stain?

Answer

Hematoxylin stains nuclei blue-black. OG-6 and the EA polychrome stain, containing eosin Y, light green SF, and Bismarck brown, colour the cytoplasm.

#303 · Cytology

Why are cytology slides labelled with pencil or solvent-resistant print on the frosted end?

Answer

Ordinary pen ink dissolves in the alcohols and xylene used during staining, so the identification would be lost. Two matching patient identifiers are required.

#304 · Cytology

What squamous cell numbers make a conventional cervical smear and a liquid-based slide adequate?

Answer

A conventional smear needs an estimated 8,000 to 12,000 well-preserved squamous cells. A liquid-based slide needs at least 5,000.

#305 · Cytology

What is a floater in cytology staining and how can it be prevented?

Answer

A clump of cells that drifts from one slide to another in a staining dish, risking a false positive. Filter and change stains regularly and separate staining lines.

#306 · Cytology

Why should sputum and other possibly tuberculous specimens be opened in a biological safety cabinet?

Answer

They may contain *Mycobacterium tuberculosis*, which spreads by aerosol. Unfixed specimens carry higher infection risk, so they are handled with containment and protective equipment.

#307 · Cytology

What temperature is normally used to hold a body fluid that cannot be processed promptly?

Answer

Refrigeration at 2 to 8 °C, or the addition of a small volume of fixative according to laboratory policy. Freezing and excess heat damage cells.

#308 · Cytology

What is the transformation zone of the cervix and why must a cervical sample include cells from it?

Answer

The area where columnar epithelium has been replaced by squamous metaplasia. Nearly all cervical squamous precancers begin here, so the sample must reach it.

#309 · Cytology

Describe the nucleus and cytoplasm of a normal superficial squamous cell.

Answer

It is a large, flat cell about 40 to 60 μm across with a small, dark, pyknotic nucleus of 4 to 6 μm and transparent pink or blue cytoplasm.

#310 · Cytology

How large is a normal intermediate cell nucleus, and how is it used as a ruler?

Answer

About 8 μm across, with fine even chromatin. A nucleus more than three times its area is considered enlarged.

#311 · Cytology

In what settings are parabasal cells a normal finding?

Answer

In children, after childbirth, and after menopause, when estrogen is low. Large numbers in a woman of reproductive age are not expected.

#312 · Cytology

How do normal endocervical cells look when seen from above and from the side?

Answer

From above they form a honeycomb of evenly spaced polygonal cells. From the side they are tall columnar cells lined up like a picket fence.

#313 · Cytology

When are benign-appearing endometrial cells considered normal, and when are they reported?

Answer

They are normal in the first 12 days of the cycle. The Bethesda System reports them in women aged 45 years or older.

#314 · Cytology

What does a maturation index of 0 / 20 / 80 mean and what hormonal state does it suggest?

Answer

0% parabasal, 20% intermediate, and 80% superficial cells. It shows strong estrogen effect, such as mid-cycle.

#315 · Cytology

What are navicular cells and when are they seen?

Answer

Boat-shaped intermediate cells with glycogen in the cytoplasm giving a yellow tint. They are typical of pregnancy and the luteal phase under progesterone influence.

#316 · Cytology

What nuclear features separate a reactive squamous cell from an abnormal one?

Answer

Reactive nuclei are only mildly enlarged with smooth membranes and fine, even chromatin. Abnormal nuclei show hyperchromasia, coarse chromatin, and irregular membranes.

#317 · Cytology

What pattern is typical of reparative change on a cervical sample?

Answer

Flat, cohesive sheets with a streaming arrangement of uniform nuclei, fine chromatin, and prominent nucleoli, with borders blending into a syncytial look.

#318 · Cytology

Why is atrophy a common source of false-positive interpretations?

Answer

Parabasal cells have a high N:C ratio and may look hyperchromatic or degenerated, and the granular debris background adds to the confusion.

#319 · Cytology

How does the N:C ratio in radiation change differ from that in malignancy?

Answer

Radiation produces enlarged cells with enlarged cytoplasm, so the N:C ratio stays normal or low. In malignancy the N:C ratio is increased.

#320 · Cytology

Which IUD-related cell change can be mistaken for adenocarcinoma?

Answer

Single or clustered glandular cells with cytoplasmic vacuoles pushing the nucleus aside, giving a signet-ring look, sometimes in tight groups with nucleoli.

#321 · Cytology

What are the three main parts of a Bethesda System cervical cytology report?

Answer

The specimen type, the adequacy of the specimen, and the interpretation or result, such as negative for intraepithelial lesion or malignancy.

#322 · Cytology

How many squamous cells are needed for a satisfactory liquid-based cervical sample?

Answer

At least 5,000 well-preserved squamous cells. A conventional smear needs an estimated 8,000 to 12,000.

#323 · Cytology

Which two viral proteins made by high-risk HPV disable the cell's tumour-suppressing controls, and which proteins do they target?

Answer

E6 promotes destruction of p53, and E7 binds the retinoblastoma protein (Rb). Together they let damaged cells keep dividing and accumulate errors that can lead to cancer.

#324 · Cytology

Which two high-risk HPV types together cause about 70% of cervical cancers?

Answer

Types 16 and 18 together cause roughly 70% of cervical cancers. Type 16 is the most common single type, and type 18 is especially linked to adenocarcinoma.

#325 · Cytology

What is the koilocyte, and which cervical cytology category does it mark?

Answer

A koilocyte is a squamous cell with a sharp perinuclear halo and an enlarged, irregular, hyperchromatic nucleus. It marks productive HPV infection and a low-grade squamous intraepithelial lesion (LSIL).

#326 · Cytology

How do LSIL and HSIL cells differ in cell size and nuclear-to-cytoplasmic ratio?

Answer

LSIL cells are mature, large cells with abundant cytoplasm and only a moderately raised N:C ratio. HSIL cells are smaller, immature cells with scant cytoplasm and a high N:C ratio.

#327 · Cytology

What size does the nucleus reach in ASC-US compared with a normal intermediate cell nucleus?

Answer

In ASC-US the nucleus is about two and a half to three times the area of a normal intermediate cell nucleus, with only a slightly raised N:C ratio.

#328 · Cytology

When is the category ASC-H used instead of ASC-US?

Answer

ASC-H is used when few small cells resemble immature metaplastic or parabasal cells with a high N:C ratio and a high-grade lesion cannot be excluded. It carries higher risk than ASC-US.

#329 · Cytology

What background finding suggests that a squamous abnormality may be invasive carcinoma?

Answer

A tumour diathesis, a granular dirty background of old blood, necrotic debris and protein, suggests invasion. Its absence does not rule cancer out.

#330 · Cytology

What does the CIN 2 and CIN 3 histology terminology correspond to in Bethesda cytology reporting?

Answer

CIN 2 and CIN 3 correspond to a high-grade squamous intraepithelial lesion (HSIL), the true precancer that requires treatment.

#331 · Cytology

What cytologic features suggest endocervical adenocarcinoma in situ (AIS)?

Answer

Crowded three-dimensional groups of columnar cells with feathering and rosette formations at the edges, elongated hyperchromatic nuclei, frequent mitoses and apoptotic bodies, and no tumour diathesis.

#332 · Cytology

Why is AGC followed up more aggressively than ASC-US?

Answer

AGC is often subtle but carries a significant risk of high-grade squamous lesion, AIS, or endocervical or endometrial cancer. Follow-up includes colposcopy and, for older or bleeding patients, endometrial evaluation.

#333 · Cytology

How is *Trichomonas vaginalis* recognized on a Papanicolaou-stained cervical sample?

Answer

It appears as a pale grey-blue to green, pear-shaped organism about the size of a neutrophil with a small eccentric nucleus, usually with many neutrophils in the background.

#334 · Cytology

What cytologic pattern identifies bacterial vaginosis on a cervical sample?

Answer

Clue cells: squamous cells covered with tiny coccobacilli so their borders look fuzzy, with few neutrophils and a loss of the usual lactobacilli.

#335 · Cytology

What three nuclear features characterize herpes simplex virus infection in cervical samples?

Answer

Multinucleation, margination of chromatin and moulding of nuclei, giving ground-glass nuclei with a thick rim. Some cells have a Cowdry type A eosinophilic inclusion with a clear halo.

#336 · Cytology

Can chlamydia be reliably diagnosed on cervical cytology, and how is it confirmed?

Answer

No. Cytology features are not specific or reliable. Chlamydia is diagnosed by a nucleic acid amplification test, not by the cytology report.

#337 · Cytology

What is the minimum squamous cell count for a satisfactory liquid-based cervical sample?

Answer

At least about 5,000 well-visualized squamous cells. A sample is also unsatisfactory when more than 75% of squamous cells are obscured by blood, inflammation or poor fixation.

#338 · Cytology

How is ASC-US managed when reflex HPV testing is used?

Answer

A positive high-risk HPV result leads to colposcopy, and a negative result returns the patient to routine screening or a repeat test at the interval set by the provincial program.

#339 · Cytology

How does a transudate differ from an exudate, and which is more useful for cytology?

Answer

A transudate results from pressure or protein imbalance (heart failure, cirrhosis) and has few cells. An exudate comes from serosal disease such as infection or cancer and is more useful for cytology.

#340 · Cytology

Why should body fluids reach the laboratory promptly, and what is done if delay is unavoidable?

Answer

Cells degenerate within hours. If delay is unavoidable, the fluid is refrigerated at about 4 degrees Celsius, and a preservative or fixative may be added.

#341 · Cytology

What is a cell block, and why is it valuable for effusion samples?

Answer

A cell block is sediment clotted or embedded and processed like tissue. It can be sectioned for special stains and immunohistochemistry, which helps identify a tumour's origin.

#342 · Cytology

What features help identify normal mesothelial cells in an effusion?

Answer

Round to oval cells with central round nuclei, smooth even chromatin, dense central cytoplasm with a paler edge, a fuzzy microvillous border, and small clear windows between neighbouring cells.

#343 · Cytology

Which cells commonly appear in reactive effusions and can mimic cancer?

Answer

Reactive mesothelial cells, which may be enlarged, binucleated or multinucleated with prominent nucleoli and three-dimensional groups. They look alike and show a gradual spectrum, unlike cancer cells.

#344 · Cytology

What three-dimensional pattern suggests metastatic adenocarcinoma in a serous fluid?

Answer

Smooth, rounded three-dimensional balls, acini or papillary groups with large irregular nuclei, prominent nucleoli and vacuolated cytoplasm, and no windows or microvillous border between cells.

#345 · Cytology

Which immunostains favour mesothelial origin, and which favour adenocarcinoma?

Answer

Calretinin, WT-1, D2-40 and CK5/6 favour mesothelial cells. MOC-31, Ber-EP4, CEA and TTF-1 (for lung) favour adenocarcinoma.

#346 · Cytology

How do lymphoma cells appear in a body fluid, and what test confirms clonality?

Answer

A monotonous population of discohesive atypical lymphoid cells with lymphoglandular bodies. Flow cytometry on fresh, unfixed fluid confirms a clonal population.

#347 · Cytology

Why must cerebrospinal fluid be processed quickly, and what does it usually test for?

Answer

CSF cells degrade rapidly, so it should be processed within about an hour. Cytology looks for leptomeningeal metastasis, lymphoma or leukaemia, and shed primary brain tumour cells.

#348 · Cytology

How is a good sputum specimen recognized under the microscope?

Answer

It contains alveolar macrophages, showing it came from the deep lung. A sample with only saliva, oral squamous cells and bacteria is inadequate.

#349 · Cytology

What is the difference between bronchial brushing and bronchoalveolar lavage?

Answer

A brush scrapes cells directly from a visible lesion and gives well-preserved cells. Lavage rinses small airways and alveoli with a larger saline volume and is often used to detect infection.

#350 · Cytology

What cytologic features identify small cell carcinoma of the lung?

Answer

Small hyperchromatic cells with scant cytoplasm, salt-and-pepper chromatin, inconspicuous nucleoli, prominent nuclear molding, chains or rosettes, and a background of necrosis and crush artifact.

#351 · Cytology

Which immunostains support lung adenocarcinoma and which support squamous carcinoma?

Answer

TTF-1 and napsin A support lung adenocarcinoma.
p40 and p63 support squamous cell carcinoma.

#352 · Cytology

Name three benign respiratory findings that must not be mistaken for tumour.

Answer

Ciliated bronchial cells, Curschmann spirals and Charcot-Leyden crystals are common benign findings. Pollen and plant fibres are typical contaminants seen in sputum.

#353 · Cytology

How is *Pneumocystis jirovecii* shown in bronchoalveolar lavage fluid?

Answer

It appears as frothy, honeycomb-like pink material on Papanicolaou stain, and the cysts are shown as round or crescent forms on a silver stain such as Grocott methenamine silver.

#354 · Cytology

What cell change is typical of cytomegalovirus infection in respiratory samples?

Answer

Very large cells with a dense intranuclear inclusion surrounded by a clear halo, giving an owl-eye appearance.

#355 · Cytology

Which type of urothelial carcinoma is the main target of the Paris System for reporting urine cytology?

Answer

High-grade urothelial carcinoma. Low-grade tumours look almost normal under the microscope, so cytology detects them poorly and the system does not depend on them.

#356 · Cytology

What nuclear-to-cytoplasmic ratio is required for a call of high-grade urothelial carcinoma in urine?

Answer

An N/C ratio of 0.7 or more, together with marked hyperchromasia and irregular chromatin and nuclear membranes, in a meaningful number of cells.

#357 · Cytology

What minimum N/C ratio and features define atypical urothelial cells under the Paris System?

Answer

An N/C ratio above 0.5 plus at least one of hyperchromasia, irregular clumped chromatin or an irregular nuclear membrane, short of the threshold for suspicious or malignant.

#358 · Cytology

Which voided urine specimen is preferred for cytology, and why is the first morning void avoided?

Answer

A second morning or later daytime void. Cells sitting in the bladder overnight degenerate, so the first morning void gives poorly preserved cells.

#359 · Cytology

Why can catheterized or bladder washing urine specimens mimic a papillary tumour?

Answer

The instrument scrapes the lining and produces tight three-dimensional urothelial cell clusters, which resemble papillary fragments but are only an artifact of collection.

#360 · Cytology

What is a decoy cell in urine, and which virus causes it?

Answer

An infected urothelial cell with an enlarged nucleus filled by a homogeneous ground-glass inclusion and margined chromatin. It is caused by BK polyomavirus, often in transplant recipients.

#361 · Cytology

How can decoy cells be confirmed and why do they matter clinically?

Answer

By SV40 large T antigen immunostaining or molecular testing. They matter because BK nephropathy can damage a transplanted kidney, and they mimic cancer cells.

#362 · Cytology

What is the classic cytomegalovirus appearance in a urine specimen?

Answer

A large cell with one huge intranuclear inclusion surrounded by a clear halo, the owl-eye appearance, sometimes with small cytoplasmic granules.

#363 · Cytology

Which cytologic features separate a breast fibroadenoma from ductal carcinoma on FNA?

Answer

Fibroadenoma shows cohesive staghorn sheets and many bare bipolar nuclei. Ductal carcinoma shows discohesive cells, pleomorphic nuclei and few or no bipolar nuclei.

#364 · Cytology

Why is invasive lobular carcinoma a common source of false-negative breast FNA results?

Answer

Its cells are small, uniform and discohesive with only mildly abnormal nuclei, so they can look benign. Intracytoplasmic lumina and single files are helpful clues.

#365 · Cytology

How many reporting categories does the IAC Yokohama System use for breast FNA, and what are they?

Answer

Five: insufficient material, benign, atypical, suspicious for malignancy and malignant.

#366 · Cytology

How is a nipple discharge specimen collected and what must happen right away?

Answer

A clean slide is touched to the nipple to make a direct smear, which is fixed immediately in alcohol-based fixative, because the cells dry quickly and lose detail.

#367 · Cytology

What are the six Bethesda System categories for thyroid FNA, in order?

Answer

Nondiagnostic, benign, atypia of undetermined significance, follicular neoplasm, suspicious for malignancy and malignant.

#368 · Cytology

What is the usual adequacy criterion for a thyroid FNA in the Bethesda System?

Answer

At least six groups of at least ten well-preserved follicular cells on a single slide, with exceptions for abundant colloid, significant atypia or thyroiditis.

#369 · Cytology

Which nuclear features establish a diagnosis of papillary thyroid carcinoma?

Answer

Enlarged overlapping nuclei with powdery, cleared chromatin, longitudinal nuclear grooves and intranuclear pseudo-inclusions, often with papillary fragments, ropy colloid and psammoma bodies.

#370 · Cytology

Why can cytology not separate follicular adenoma from follicular carcinoma, and what is the consequence?

Answer

Malignancy is proven only by capsular or vascular invasion in the removed nodule. The FNA is reported as follicular neoplasm and the patient goes to surgery.

#371 · Cytology

What is the difference between the aspiration and the capillary (non-aspiration) FNA methods?

Answer

Aspiration applies syringe suction while the needle moves in the lesion. The capillary method uses no suction, so cells enter the needle by capillary action, giving less blood dilution in vascular organs.

#372 · Cytology

What does an air-dried Romanowsky smear show best compared with an alcohol-fixed Pap smear?

Answer

Air-dried smears show cytoplasm, mucin, colloid and stroma, and cells look larger. Alcohol-fixed Pap smears show fine nuclear detail such as chromatin and nucleoli.

#373 · Cytology

What is air-drying artifact and how is it prevented on Pap smears?

Answer

Cells swell and nuclei turn pale and blurred when a smear dries before fixation. It is prevented by placing the slide into alcohol or spraying fixative immediately.

#374 · Cytology

Why are FNA slides labelled in pencil on the frosted end?

Answer

Pencil does not dissolve in alcohol fixative or stains, so two patient identifiers stay readable through fixation and staining and specimen mix-ups are avoided.

#375 · Cytology

What are the main goals of rapid on-site evaluation (ROSE)?

Answer

Judge adequacy, decide if more passes are needed, triage material for flow cytometry, culture or cell block, and guide the operator, not usually to give a final diagnosis.

#376 · Cytology

Why is a cell block valuable in FNA cytology?

Answer

It preserves cell architecture and provides many unstained slides for immunostains and molecular testing, turning a descriptive result into a more definite diagnosis.

#377 · Cytology

Which medium is used for a suspected lymphoma FNA specimen meant for flow cytometry?

Answer

A balanced culture medium such as RPMI, because flow cytometry needs living cells with intact surface proteins. Alcohol and formalin destroy this.

#378 · Cytology

How does a reactive lymph node differ from lymphoma on FNA smears?

Answer

A reactive node shows a polymorphous mix of small lymphocytes, larger cells and tingible-body macrophages. Lymphoma shows a monotonous atypical population and needs flow cytometry.

#379 · Cytology

What is the classic stromal appearance of pleomorphic adenoma of the salivary gland on Romanowsky stain?

Answer

Bright magenta, fibrillar chondromyxoid stroma with bland epithelial and myoepithelial cells embedded in it.

#380 · Cytology

Which cytologic features suggest pancreatic ductal adenocarcinoma on EUS-guided FNA?

Answer

Crowded sheets with a drunken honeycomb pattern, loss of polarity, irregular nuclear contours and nuclear size varying about fourfold within a sheet.

#381 · Cytology

Why must a pheochromocytoma be considered before adrenal FNA?

Answer

It releases catecholamines, so puncture without preparation can cause a hypertensive crisis. Clinical and imaging information must accompany the request.

#382 · Cytology

Which sites does EBUS-TBNA sample and what are its common uses?

Answer

Mediastinal and hilar lymph nodes, used to stage lung cancer, diagnose sarcoidosis and sample lymphoma. The lymph node station is recorded on each specimen.

#383 · Cytology

Which cytologic features characterize small cell carcinoma?

Answer

Small cells with scant cytoplasm, nuclear moulding, fine salt-and-pepper chromatin, crush artifact, frequent necrosis and many mitoses.

#384 · Cytology

Why is CSF cytology processed promptly?

Answer

CSF has few cells and they begin to degrade within about an hour, so delay loses malignant or leukemic cells that could be seen by cytocentrifugation.

#385 · Cytology

What is the safe handling rule for used FNA needles?

Answer

Never recap with two hands and never pass a needle by hand. Place it directly in a puncture-resistant sharps container at the point of use.

#386 · Cytology

What does immunocytochemistry detect in cells, and how is the result made visible?

Answer

It detects specific proteins (antigens) using antibodies. A detection system and a chromogen such as DAB leave a coloured deposit, often brown, where the antigen is present.

#387 · Cytology

Why does every immunocytochemistry run need a positive and a negative control?

Answer

The positive control proves the stain worked, and the negative control shows background staining. If the positive control fails, the whole run is invalid regardless of how the patient cells look.

#388 · Cytology

Which specimen type is preferred for immunocytochemistry and molecular testing on cytology material, and why?

Answer

The cell block, because it is paraffin embedded and behaves like tissue, allowing many stains and molecular tests from one sample.

#389 · Cytology

Name markers that point toward mesothelium versus adenocarcinoma in an effusion.

Answer

Calretinin and WT1 point toward mesothelial cells. MOC-31, BerEP4 and claudin-4 point toward adenocarcinoma.

#390 · Cytology

How is sample adequacy judged for molecular testing on a cytology sample?

Answer

By tumour cell content, meaning the percentage and number of tumour cells, not just the presence of cells. Too few tumour cells can cause a false negative result.

#391 · Cytology

Why should strong acid decalcifying agents be avoided on samples meant for molecular testing?

Answer

Strong acids damage and fragment DNA, which can make the molecular assay fail or give unreliable results.

#392 · Cytology

What is cotesting in cervical screening?

Answer

Cotesting means performing cytology and HPV testing on the same cervical sample at the same visit.

#393 · Cytology

Which HPV types are separated by genotyping because they carry the highest risk?

Answer

Types 16 and 18 are genotyped separately, since they cause most cervical cancers and can lead directly to colposcopy.

#394 · Cytology

What should a technologist do when the internal control of an HPV assay is invalid?

Answer

Do not report the sample as negative. Report it as unsatisfactory or repeat the test, because the invalid control means the assay may not have worked.

#395 · Cytology

How do laboratories prevent contamination in amplification-based HPV testing?

Answer

They use separate pre-amplification and post-amplification work areas, change gloves often, and run internal, positive and negative controls.

#396 · Cytology

What does flow cytometry show that suggests lymphoma rather than a reactive lymphocyte population?

Answer

Clonality: a skewed kappa to lambda light chain ratio or an abnormal marker pattern, since a lymphoma arises from a single cell.

#397 · Cytology

Why must fluids for flow cytometry stay unfixed?

Answer

Alcohol or formalin destroys many surface antigens and kills cells, so flow cytometry needs a fresh specimen in suitable medium and prompt transport.

#398 · Cytology

What is a commonly cited upper limit for manual gynaecologic screening workload, and why is it set?

Answer

About 100 slides in a 24-hour period, limited to reduce fatigue and missed abnormal cells. Local requirements should be followed.

#399 · Cytology

What is the purpose of random rescreening of negative slides, and who performs it?

Answer

It estimates how often abnormal cells are missed. A commonly used rate is 10% of negatives, and a different qualified person performs it.

#400 · Cytology

In cytohistologic correlation, what three types of error may explain a discrepancy?

Answer

Sampling error (abnormal cells not on the slide), screening error (cells present but not found) and interpretation error (cells seen but called incorrectly).

#401 · Cytogenetics

Why can chromosomes only be studied in living, dividing cells?

Answer

Chromosomes are clearly visible only in metaphase, a short stage of cell division, so cells must be alive and dividing when the culture is harvested.

#402 · Cytogenetics

Which anticoagulant is used for cytogenetic blood and bone marrow specimens, and which must be avoided?

Answer

Sodium heparin is used. EDTA and citrate damage cell growth and are unacceptable for culture, so those specimens may need recollection.

#403 · Cytogenetics

Why is formalin never used for cytogenetic specimens?

Answer

Formalin kills the cells and prevents culture, so tissue for chromosome analysis goes into sterile culture medium or saline.

#404 · Cytogenetics

At what gestational age are amniocentesis and CVS usually performed?

Answer

Amniocentesis is usually done at about 15 to 20 weeks, and chorionic villus sampling at about 10 to 13 weeks.

#405 · Cytogenetics

What are the two parts of CVS tissue, and how does each get analysed?

Answer

The cytotrophoblast divides quickly and can be studied directly. The mesenchymal core is grown in culture to confirm results.

#406 · Cytogenetics

How should cytogenetic specimens usually be transported?

Answer

At room temperature, reaching the laboratory the same day. Freezing and extreme heat kill the cells, and refrigeration overnight is avoided unless instructed.

#407 · Cytogenetics

Why is all cytogenetic culture work done in a biological safety cabinet?

Answer

It protects the culture from bacteria and fungi, which outgrow human cells, and it protects the worker from infectious specimens.

#408 · Cytogenetics

What is PHA and which cells does it stimulate in blood culture?

Answer

Phytohemagglutinin is a mitogen that pushes cells to divide. It stimulates T lymphocytes, and blood cultures usually grow for about 72 hours.

#409 · Cytogenetics

Why is bone marrow often set up as direct, overnight and 48-hour cultures?

Answer

Different cancer cells divide at different speeds, so several timings improve the chance of capturing the abnormal cells in metaphase.

#410 · Cytogenetics

What does colcemid do in chromosome harvest?

Answer

It blocks spindle fibre formation so dividing cells stay stuck in metaphase, increasing the number of metaphases for analysis.

#411 · Cytogenetics

What happens if colcemid exposure is too long?

Answer

Chromosomes become very short and over-condensed, making bands hard to read, and cells can die.

#412 · Cytogenetics

What is the purpose of the hypotonic solution, and which reagent is commonly used?

Answer

0.075 molar potassium chloride makes cells swell so chromosomes spread apart without overlapping, and it also breaks up red cells.

#413 · Cytogenetics

What is the standard cytogenetic fixative and how is it handled?

Answer

Three parts methanol to one part glacial acetic acid, made fresh and added slowly to avoid clumping cells. Old fixative spreads poorly.

#414 · Cytogenetics

How does G-banding work?

Answer

Trypsin partly digests chromosomal proteins, then Giemsa stain produces a dark and light band pattern unique to each chromosome.

#415 · Cytogenetics

What is a typical cause of fuzzy chromosomes versus pale ghost-like chromosomes after G-banding?

Answer

Too little trypsin gives fuzzy, uniformly dark chromosomes. Too much trypsin gives pale, ghost-like chromosomes with thin outlines.

#416 · Cytogenetics

If several unrelated cultures fail on the same day, what is the most likely cause?

Answer

A reagent or equipment problem, such as bad medium, inactive mitogen or incubator fault, whereas a single failed case suggests a specimen problem.

#417 · Cytogenetics

Why must maternal cell contamination be considered in prenatal cytogenetics?

Answer

Maternal cells may mix with fetal cells and cause a falsely reassuring normal result, so multiple cultures or molecular checks confirm fetal origin.

#418 · Cytogenetics

How many chromosomes does a normal human cell have, and how is a normal female and male karyotype written in ISCN?

Answer

A normal cell has 46 chromosomes: 22 pairs of autosomes plus two sex chromosomes. A normal female is written 46,XX and a normal male is written 46,XY.

#419 · Cytogenetics

Which chromosomes are acrocentric, and why do they matter in cytogenetics?

Answer

Chromosomes 13, 14, 15, 21 and 22 are acrocentric. Their short arms carry satellites and take part in Robertsonian translocations, which are fusions at the centromere with loss of the short arms.

#420 · Cytogenetics

What does the p arm and the q arm of a chromosome refer to?

Answer

The p arm is the short arm, named from the French *petit*, and the q arm is the long arm. The centromere sits between them.

#421 · Cytogenetics

What stain and pretreatment produce standard G-banding, and what do the dark bands represent?

Answer

Chromosomes are treated with trypsin and stained with Giemsa. Dark G-bands are AT-rich and contain fewer active genes, while light bands are GC-rich and gene-rich.

#422 · Cytogenetics

Which spindle inhibitor is used to arrest cells in metaphase for a karyotype?

Answer

Colcemid arrests dividing cells in metaphase by blocking the spindle. The cells are then treated with a hypotonic solution, fixed, dropped on slides and banded.

#423 · Cytogenetics

What are the three autosomal trisomies commonly seen in liveborn infants?

Answer

Trisomy 21 (Down syndrome), trisomy 18 (Edwards syndrome) and trisomy 13 (Patau syndrome). Most other autosomal trisomies and monosomies end in early pregnancy loss.

#424 · Cytogenetics

What is nondisjunction, and what maternal factor raises its risk?

Answer

Nondisjunction is failure of chromosomes or sister chromatids to separate properly during cell division, giving gains or losses. Advancing maternal age raises the risk of meiotic nondisjunction.

#425 · Cytogenetics

How many chromosomes are in triploidy and tetraploidy?

Answer

Triploidy has 69 chromosomes and tetraploidy has 92. Triploidy is a common cause of miscarriage, while tetraploid cells in a culture are often a culture artifact.

#426 · Cytogenetics

What is a Robertsonian translocation carrier's chromosome count, and what is the clinical concern?

Answer

A carrier typically has 45 chromosomes but is usually healthy. The concern is unbalanced gametes, for example a chromosome 21 translocation that can cause translocation Down syndrome in a child.

#427 · Cytogenetics

What is the difference between a paracentric and a pericentric inversion?

Answer

A paracentric inversion lies within one arm and does not include the centromere. A pericentric inversion includes the centromere and spans both arms.

#428 · Cytogenetics

How is Turner syndrome written in ISCN, and what is Klinefelter syndrome's karyotype?

Answer

Turner syndrome is most commonly 45,X in a female. Klinefelter syndrome is 47,XXY in a male, usually with tall stature, small testes and infertility.

#429 · Cytogenetics

What is the difference between mosaicism and chimerism?

Answer

Mosaicism is two or more cell lines from a single zygote, while chimerism is cell lines from two different zygotes, for example after a marrow transplant or fusion of two embryos.

#430 · Cytogenetics

What is pseudomosaicism, and how is it usually recognized?

Answer

Pseudomosaicism is an abnormality that arose in culture and is not in the patient. It is usually confined to a single culture or colony, or a few cells, rather than several independent cultures.

#431 · Cytogenetics

What is confined placental mosaicism and why does it matter in chorionic villus sampling?

Answer

An abnormal cell line is present in the placenta but not the fetus. A mosaic villus result often needs follow-up amniocentesis to learn whether the fetus itself is affected.

#432 · Cytogenetics

What is the Philadelphia chromosome and what does it create?

Answer

The Philadelphia chromosome is $t(9;22)(q34;q11.2)$. It joins ABL1 to BCR, creating the BCR::ABL1 fusion, seen in almost all chronic myeloid leukemia and guiding tyrosine kinase inhibitor therapy.

#433 · Cytogenetics

What is the clonality rule for a cancer karyotype?

Answer

A clone requires at least two cells with the same structural abnormality or chromosome gain, or at least three cells with the same chromosome loss.

#434 · Cytogenetics

Which recurrent translocation is typical of acute promyelocytic leukemia, and which drug class targets it?

Answer

The t(15;17)(q24;q21) joins PML and RARA. Acute promyelocytic leukemia responds to all-trans retinoic acid, so early recognition has direct treatment consequences.

#435 · Cytogenetics

What are the main steps of a FISH procedure in order?

Answer

Specimen fixation and pretreatment, denaturation of probe and target DNA, hybridization, post-hybridization stringency washes, DAPI counterstain, and viewing under a fluorescence microscope.

#436 · Cytogenetics

What three factors control the stringency of a post-hybridization FISH wash?

Answer

Temperature, salt concentration and time. A wash that is too stringent strips true signal, while one that is too lenient leaves background that makes scoring hard.

#437 · Cytogenetics

What does a centromeric enumeration probe detect, and why is its signal bright?

Answer

It counts copies of a specific chromosome by binding repeated alpha-satellite DNA at the centromere. The repeated target gives a bright, reliable signal, useful for trisomy detection in interphase.

#438 · Cytogenetics

What is a locus-specific FISH probe used for?

Answer

It binds one gene or small unique region to detect a deletion, duplication or amplification. A normal nucleus shows two signals and a deletion shows only one.

#439 · Cytogenetics

When is a whole chromosome paint used, and why does it not work well in interphase?

Answer

It is used on metaphase cells to identify the origin of extra material or confirm a complex rearrangement. In interphase, chromosomes are not condensed, so the signal is diffuse and uninformative.

#440 · Cytogenetics

What is the typical abnormal signal pattern with a BCR::ABL1 dual-colour, dual-fusion probe?

Answer

One red, one green and two fusion signals per nucleus. A normal nucleus shows two red and two green signals with no fusion signal.

#441 · Cytogenetics

What is the advantage of a break-apart probe over a dual-fusion probe?

Answer

It detects rearrangement of a gene even when the partner gene is unknown or varies. A normal gene shows fused signals, and a rearranged gene shows separated colours.

#442 · Cytogenetics

How is a FISH normal cutoff value determined?

Answer

During validation, normal control specimens are scored to find the false-positive rate for each probe. The cutoff is usually set statistically, for example as the mean of controls plus three standard deviations.

#443 · Cytogenetics

What are the main advantages and limits of interphase FISH compared with metaphase FISH?

Answer

Interphase FISH needs no culture, gives rapid results and allows scoring of many nuclei. It cannot show which chromosome carries the signal or reveal abnormalities outside the probe target.

#444 · Cytogenetics

How does aCGH detect a copy number gain or loss?

Answer

Patient DNA and reference DNA are labelled with different dyes and hybridized to array probes. A gain gives more patient colour, and a loss gives more reference colour, seen in the log ratio.

#445 · Cytogenetics

What can a SNP array detect that aCGH cannot?

Answer

It measures allele patterns as well as copy number, so it can detect long regions of homozygosity, uniparental disomy and triploidy, which do not change the aCGH ratio.

#446 · Cytogenetics

Which types of chromosome abnormality can a chromosomal microarray not detect?

Answer

Balanced rearrangements such as balanced translocations and inversions, because the total DNA amount does not change. Low-level mosaicism may also be missed.

#447 · Cytogenetics

What five categories are used to classify a copy number variant?

Answer

Pathogenic, likely pathogenic, variant of uncertain significance, likely benign and benign. Evidence includes size, gene content, known syndromes, population databases and inheritance.

#448 · Cytogenetics

Why must the genome build be reported with microarray coordinates?

Answer

Genomic coordinates change between builds such as GRCh37 and GRCh38, so the same coordinates can point to different locations. Reporting the build keeps results interpretable and comparable.

#449 · Cytogenetics

How does QF-PCR detect trisomy, and what else can it reveal in prenatal samples?

Answer

It amplifies short tandem repeat markers and measures fluorescent peaks. A trisomy shows three peaks or a 2 to 1 ratio, and maternal cell contamination shows as extra maternal peaks.

#450 · Cytogenetics

Why must formamide used in FISH be handled with particular care?

Answer

Formamide is a reproductive hazard. It should be handled in a fume hood with gloves, eye protection and a lab coat, and disposed of as hazardous waste.

#451 · Molecular Diagnostics

What is the main purpose of the extraction step in molecular testing of a patient specimen?

Answer

It frees nucleic acid from cells and separates it from proteins, lipids and salts that could interfere with the test.

#452 · Molecular Diagnostics

Which anticoagulant is preferred for blood used in most DNA and RNA testing, and which one inhibits PCR?

Answer

EDTA is preferred. Heparin inhibits PCR because it binds to the DNA polymerase, so heparinized tubes are generally rejected.

#453 · Molecular Diagnostics

Why must specimens for RNA testing be kept cold and processed promptly?

Answer

RNases are found everywhere and rapidly destroy RNA, so RNA is much less stable than DNA and needs cold storage or preservative.

#454 · Molecular Diagnostics

What are the four basic steps common to all nucleic acid extraction methods?

Answer

Lyse the cells, bind or separate the nucleic acid, wash away contaminants, and elute the pure nucleic acid in a small volume.

#455 · Molecular Diagnostics

Why are magnetic bead extraction methods the basis of most automated extractors?

Answer

Beads hold the nucleic acid while a magnet moves them for washing, so no centrifugation is needed and the system can stay closed.

#456 · Molecular Diagnostics

What A260/A280 ratio is expected for pure DNA and for pure RNA?

Answer

Pure DNA gives about 1.8 and pure RNA gives about 2.0. Lower values suggest protein or phenol carry-over.

#457 · Molecular Diagnostics

What does a low A260/A230 ratio suggest about an extracted sample?

Answer

It suggests carry-over of salts, carbohydrates or organic solvents such as guanidine or ethanol, which may inhibit downstream reactions.

#458 · Molecular Diagnostics

Why is fluorometric quantification often preferred over spectrophotometry before sequencing library preparation?

Answer

Fluorescent dyes bind only double-stranded DNA or RNA, so they are more specific and sensitive than absorbance, which also reads free nucleotides.

#459 · Molecular Diagnostics

What is the main rule of unidirectional workflow in a PCR laboratory?

Answer

Work moves only from clean pre-PCR areas toward post-PCR areas, and people, equipment and materials do not move backward.

#460 · Molecular Diagnostics

What is amplicon carry-over and how is it prevented?

Answer

It is contamination of new reactions by earlier amplified product. It is prevented by separated areas, filter tips, closed tubes and UNG treatment.

#461 · Molecular Diagnostics

How does UNG (uracil-DNA glycosylase) reduce PCR contamination?

Answer

With dUTP used in place of dTTP, UNG destroys uracil-containing product from earlier runs before new amplification begins.

#462 · Molecular Diagnostics

What does amplification in the no-template control mean?

Answer

It signals contamination, so the run is invalid and the source must be investigated and the area decontaminated as documented.

#463 · Molecular Diagnostics

What happens at each of the three steps of a PCR cycle?

Answer

Denaturation near 95 °C separates the strands, annealing lets primers bind, and extension near 72 °C lets polymerase build new strands.

#464 · Molecular Diagnostics

What are the components of a standard PCR reaction mixture?

Answer

Template DNA, two primers, heat-stable DNA polymerase such as Taq, dNTPs, magnesium chloride and buffer.

#465 · Molecular Diagnostics

What is the effect of setting the annealing temperature too low?

Answer

Primers bind non-specifically, producing extra products and false signals. Too high a temperature reduces yield or gives no product.

#466 · Molecular Diagnostics

What is a primer-dimer and why is it a problem in dye-based real-time PCR?

Answer

Primers anneal to each other and are extended. They consume reagents and give fluorescent signal that can be mistaken for target.

#467 · Molecular Diagnostics

What does the Ct value represent in real-time PCR?

Answer

The cycle at which fluorescence crosses the threshold in the exponential phase. A lower Ct means more starting target.

#468 · Molecular Diagnostics

At ideal efficiency, how many cycles separate a ten-fold difference in starting target?

Answer

About 3.3 cycles, since each cycle doubles the product. The standard curve slope is then near -3.32.

#469 · Molecular Diagnostics

What is the purpose of an internal or extraction control in each specimen?

Answer

It shows that extraction and amplification worked in that specimen. If it fails, inhibition or extraction failure is possible and a negative cannot be reported.

#470 · Molecular Diagnostics

How does reverse transcription PCR differ from standard PCR?

Answer

Reverse transcriptase first makes cDNA from RNA, and then PCR amplifies the cDNA, so it can detect RNA viruses and transcripts.

#471 · Molecular Diagnostics

How does digital PCR quantify a target without a standard curve?

Answer

The reaction is split into many partitions, positive ones are counted after amplification, and Poisson statistics give an absolute concentration.

#472 · Molecular Diagnostics

What does a melt curve show after dye-based real-time PCR?

Answer

A single sharp peak suggests one specific product, and an extra lower-temperature peak often indicates primer-dimers or non-specific product.

#473 · Molecular Diagnostics

What are common PCR inhibitors and what is the usual response when inhibition is found?

Answer

Heparin, heme, bile salts, alcohol residue and swab materials. Dilute the extract, re-extract or clean up, then repeat the test.

#474 · Molecular Diagnostics

In gel electrophoresis, why do smaller DNA fragments travel farther?

Answer

DNA is negatively charged and moves toward the anode, and smaller fragments pass through the gel pores more easily than larger ones.

#475 · Molecular Diagnostics

What is the difference between verification and validation of a molecular assay?

Answer

Verification confirms a laboratory can achieve an approved assay's stated performance. Validation establishes performance of a laboratory-developed or modified test.

#476 · Molecular Diagnostics

How does the Sanger sequencing reaction determine the order of bases?

Answer

Fluorescent ddNTPs stop strand growth at random positions, and capillary electrophoresis reads the terminal dye of each fragment by size.

#477 · Molecular Diagnostics

How many bases does a Sanger read usually cover with high accuracy?

Answer

Roughly 500 to 1,000 bases in one region, which is why it suits confirmation of variants and short targets.

#478 · Molecular Diagnostics

Why can Sanger sequencing miss a somatic variant present in a small fraction of cells?

Answer

The mutant peak is lost in the background of the normal peak when the variant is below roughly 15 to 20 percent of the sample.

#479 · Molecular Diagnostics

What does a heterozygous position look like on a Sanger electropherogram?

Answer

Two overlapping peaks of different colours at one position, each about half the height of a normal single peak.

#480 · Molecular Diagnostics

What is the purpose of adapters and barcodes in NGS library preparation?

Answer

Adapters let fragments bind to the sequencer, and barcodes allow many patient libraries to be pooled and sorted back to the right patient.

#481 · Molecular Diagnostics

How do hybrid capture and amplicon enrichment differ for targeted NGS?

Answer

Hybrid capture uses probes to pull out targets and gives more even coverage. Amplicon methods use PCR primers and need less input but can miss variants under primers.

#482 · Molecular Diagnostics

What is sequencing depth and what is coverage in NGS?

Answer

Depth is the number of reads covering a position. Coverage is the share of the target covered at or above a minimum depth.

#483 · Molecular Diagnostics

What is variant allele fraction and what is expected for a germline heterozygous variant?

Answer

It is the proportion of reads with the variant. A germline heterozygous variant is expected near 50 percent.

#484 · Molecular Diagnostics

What is a frameshift variant?

Answer

An insertion or deletion whose length is not a multiple of three, which shifts the reading frame and often destroys the protein.

#485 · Molecular Diagnostics

Which methods are better than short-read panels for detecting copy number variants and fusions?

Answer

Depth analysis, microarray or MLPA suit copy number changes, and RNA sequencing, FISH or RT-PCR suit fusion genes.

#486 · Molecular Diagnostics

How do gene panels, exome and genome sequencing differ in scope?

Answer

Panels target selected genes with deep coverage, exome covers coding regions, and genome covers coding and non-coding DNA with the most data.

#487 · Molecular Diagnostics

What are the five tiers of variant classification?

Answer

Pathogenic, likely pathogenic, variant of uncertain significance, likely benign and benign, following ACMG and AMP guidelines.

#488 · Molecular Diagnostics

What does a variant of uncertain significance mean for clinical care?

Answer

Evidence is insufficient or conflicting. It is not a diagnosis and should not drive clinical decisions, but it may be reclassified later.

#489 · Molecular Diagnostics

Which kinds of evidence support classification of a variant?

Answer

Population frequency, computational predictions, functional studies, segregation in families, de novo occurrence and previous reports in curated databases.

#490 · Molecular Diagnostics

What is the key difference between germline and somatic testing?

Answer

Germline variants are in every cell and heritable, while somatic variants are acquired in tumour cells only and are not inherited.

#491 · Molecular Diagnostics

Why does tumour cell content of a specimen matter in somatic testing?

Answer

A low proportion of tumour cells lowers the variant allele fraction and can cause a false negative, so pathologists estimate content and macrodissect.

#492 · Molecular Diagnostics

What does a somatic result with a VAF near 50 percent in a hereditary cancer gene suggest?

Answer

It may reflect a germline variant, so the laboratory follows policy to recommend confirmatory germline testing.

#493 · Molecular Diagnostics

What is minimal residual disease and why is it measured?

Answer

It is the small number of cancer cells remaining after treatment. Measuring it judges response, predicts relapse and guides treatment decisions.

#494 · Molecular Diagnostics

How is chronic myeloid leukemia response monitored molecularly?

Answer

By serial quantitative RT-PCR for BCR-ABL1 transcripts, reported on the international scale, with a rise prompting ABL1 mutation testing.

#495 · Molecular Diagnostics

Which genes are examples of pharmacogenomic testing and what do they predict?

Answer

CYP2C19 for clopidogrel activation, TPMT and NUDT15 for thiopurine toxicity, DPYD for fluoropyrimidine toxicity, and HLA-B for hypersensitivity.

#496 · Molecular Diagnostics

What alteration defines chronic myeloid leukemia?

Answer

The BCR-ABL1 fusion from the 9;22 translocation, or Philadelphia chromosome, which produces a constitutively active tyrosine kinase.

#497 · Molecular Diagnostics

What is JAK2 V617F and which diseases is it associated with?

Answer

A point mutation in JAK2 found in most polycythemia vera and about half of essential thrombocythemia and primary myelofibrosis cases.

#498 · Molecular Diagnostics

What are BRAF V600E and common EGFR mutations used for in cancer testing?

Answer

BRAF V600E predicts response to BRAF inhibitors in melanoma. EGFR exon 19 deletions and L858R predict response to EGFR inhibitors in lung cancer.

#499 · Molecular Diagnostics

What does a Phred quality score of 30 mean?

Answer

An estimated 1 in 1,000 chance that the base call is wrong, or about 99.9 percent base call accuracy.

#500 · Molecular Diagnostics

What must a molecular sequencing report include beyond the variant itself?

Answer

Patient and specimen identifiers, method and its limits, standard nomenclature, classification, interpretation, follow-up advice and limitations, with critical results communicated directly.